Balanced Fair Division and Generalized Kneser Graphs

Sharareh Alipour
New Uzbekistan University
Tashkent, Uzbekistan

Amir Jafari
New Uzbekistan University
Tashkent, Uzbekistan

Abstract

We study the allocation of indivisible goods between two prescribed nonempty groups whose members have arbitrary monotone valuations. We prove that at most five agents always admit an allocation that is envy-free up to one good (EF1), with bundle sizes differing by at most one. This balanced guarantee is sharp: every split of six agents can fail. We also guarantee balanced allocations that are envy-free up to two goods (EF2) for at most nine agents, improving the previous five-agent consequence of known chromatic bounds. Both existence guarantees hold for any number of goods.

Building on the connection established by Kyropoulou, Suksompong, and Voudouris, we derive these guarantees from chromatic bounds for generalized Kneser graphs. The graph $\mathrm{KG}(2n, n, r)$ has the n-subsets of a $2n$-set as vertices, with edges joining sets whose intersection has size less than r. We prove that its chromatic number is six for $r = 2$ and $n \geq 2$, resolving the conjecture of Jafari and Alipour, and at least ten for $r = 3$ and $n \geq 4$, improving the previous lower bound of six; the known upper bound is fourteen.

We show that the standard box-complex $\mathbb{Z}_2$-index bound yields exactly five and nine colors for $n \geq 3$ and $n \geq 5$, respectively. To obtain the extra color, we use restrictions on maps induced by actual colorings. A hypothetical proper five- or nine-coloring must induce a nonzero map on homology. We exhibit generators supported on pentagon or rook configurations and show that local coloring constraints force every generator to map to zero. The ten-color proof compares integral and fractional common centers and uses a sharp rounding theorem for five subset constraints.

A linear-time retraction from the full allocation graph onto its balanced layers adds cardinality balance to known nonlinear discrepancy guarantees. For each fixed removal allowance, in the guaranteed-existence range, finding a balanced fair allocation or an explicit monotonicity violation from valuation circuits is polynomial-time equivalent to finding a monochromatic edge in a circuit-specified coloring of the corresponding graph. The reverse reduction produces normalized monotone binary valuations using ordinary circuits.

Contents

1 Introduction 1

2 Balanced fair division and the chromatic threshold 3

3 The six-color theorem 5

4 The ten-color lower bound 8

5 Balancing the full allocation graph 10

6 Equivalent search problems 12

7 The search reductions 13

8 The local rook coloring obstruction 14

9 The common topological and geometric tools 16

10 The uniform box-complex index bound 23

11 Integral and fractional radius-two centers 26

A Filling from fibers 32

B The nine-point complex 33

C The eight-point complex 36

1 Introduction

Suppose that indivisible goods must be divided between two prescribed nonempty groups, such as households sharing their allocated goods. Every member of a group receives the same bundle, but each agent has her own monotone valuation. Exact envy-freeness may be impossible even for one good. An allocation is *envy-free up to c goods*, or EFc, if each agent ceases to envy the other bundle after at most c goods are removed from it; the removed goods may depend on the agent. We require also that the two bundle sizes differ by at most one, and call such an allocation *balanced*. Our first result determines the number of agents for which balanced EF1 is always possible and improves the corresponding guarantee for EF2.

Theorem 1.1 (Sharp balanced EF1; EF2 for nine agents)**.** *Every instance with at most five agents admits balanced EF1, and every instance with at most nine agents admits balanced EF2, for arbitrary monotone valuations and every prescribed split into two nonempty groups. For every split of six agents and every even number of goods at least four, monotone* 0/1*-valued valuations can prevent balanced EF1.*

Kyropoulou, Suksompong, and Voudouris related balanced EF1 to a generalized Kneser graph and proved matching allocation obstructions [KSV20, Theorem 4.4 and its tightness argument]. They observed that a conjecture of Jafari and Alipour [JA17] would guarantee five agents. We prove that conjecture, making their sharp threshold unconditional for both splits $(1, 4)$ and $(2, 3)$. The result also extends the four-agent additive guarantee of Gölz and Yaghoubizade [GY26] to five agents with arbitrary monotone valuations. For EF2, the previous chromatic bound of Daneshpajouh [Dan19] guaranteed five agents; ours guarantees nine. Section 2 states and proves the exact correspondence for every allowance c.

The chromatic results. Let $\mathrm{KG}(N, k, r)$ have the k-subsets of $[N] = \{1, \dots, N\}$ as vertices; two distinct sets are adjacent when their intersection has size less than r. A proper coloring partitions these sets into families in which every pair has at least r common elements. Write $\chi(G)$ for the chromatic number and put $G_{n,r} = \mathrm{KG}(2n, n, r)$. For $r = 1$, the balanced graph consists of complementary pairs and has chromatic number two.

Theorem 1.2. *For every $n \geq 2$, $\chi(G_{n,2}) = 6$. For every $n \geq 4$,*

$$10 \leq \chi(G_{n,3}) \leq 14.$$

In the second statement, equality to fourteen holds when $n = 4$.

The six-color equality is the conjecture of [JA17] from 2017. The ten-color bound raises the uniform lower bound from six [Dan19, Theorem 1]; fourteen is the Fano-plane upper bound of Balogh, Cherkashin, and Kiselev [BCK19]. These are also allocation thresholds: for $2n$ goods with $n > c$, balanced EFc is guaranteed for every profile of monotone valuations and a prescribed split of q agents exactly when $q < \chi(G_{n,c+1})$. A rejecting agent gives an independent family of allocations, and every independent family can be realized by a binary monotone valuation. Thus the correspondence gives both existence and counterexamples.

We also construct a linear-time balancing retraction and extend the correspondence to allocation and coloring search.

The extra color beyond the index bound. Lovász's topological proof of Kneser's conjecture turns graph colorings into maps to spheres [Lov78]. In its box-complex form, $B(G)$ records complete bipartite subgraphs, with an involution exchanging their sides. Its $\mathbb{Z}_2$-index is the least d for which a continuous map $B(G) \to S^d$ sends this exchange to $x \mapsto -x$. The standard bound is $\chi(G) \geq \operatorname{ind}_{\mathbb{Z}_2} B(G) + 2$.

Theorem 1.3. *For $r \geq 2$ and $n \geq 2r - 1$,*

$$\operatorname{ind}_{\mathbb{Z}_2} B(G_{n,r}) \leq 4r - 5.$$

Equality holds for $r = 2$ and $r = 3$.

The exact indices are therefore three and seven, giving only five and nine colors when $n \geq 3$ and $n \geq 5$, respectively. Thus the standard index bound stops short of both chromatic conclusions. Our argument uses the additional restrictions on maps induced by actual colorings. A putative five- or nine-coloring must induce a nonzero map on homology. We give generators supported on pentagon or rook configurations and prove that every such configuration repeats a color, forcing every generator to map to zero. The local coloring arguments are a count on five points for the pentagon and a sequence of forced colors for the rook graph.

Identifying the generators is the topological part of the proof. We first allow the common centers of a family of sets to be fractional. The resulting complex has a geometric model made of coordinate faces of a hypersimplex, a slice of a cube with fixed coordinate sum. Boundaries of the first omitted faces generate its relevant homology; the pentagon and rook cycles map with degree one onto these boundaries. For radius one, rounding five labels preserves the chain groups and boundary maps that compute third homology.

For radius two, the relevant seven-dimensional cycles require control of eight-dimensional fillings, which five-label rounding does not provide. We construct a chain map that returns fractional fillings to integral ones while fixing the original cycle. The local extension steps reduce to integral homology calculations in two fixed complexes with 84 and 70 labels. This comparison, proved in Section 11, is what permits the same coloring argument to reach ten colors.

General allowances and computation. The full allocation graph allows bundles of all sizes. We construct a complement-preserving retraction onto its central layers, in linear time, and prove that its chromatic number equals that of the balanced graph (Theorem 5.1). Combining this with the nonlinear discrepancy bounds obtained independently and concurrently in [HMMS26, DF25] gives balanced EFc for q agents with $c = O(\sqrt{q \log q})$. The discrepancy estimate is inherited; the retraction supplies balance.

For fixed c, the correspondence also extends to search with succinct inputs when $n > c$ and $q < \chi(G_{n,c+1})$ (Theorem 6.2). Given rational valuation circuits, finding a balanced EFc allocation or an explicit monotonicity violation is polynomial-time equivalent to finding a monochromatic edge in a circuit-specified q-coloring of $G_{n,c+1}$. The reductions preserve the goods, allowance, and number of agents. The reverse reduction produces normalized monotone binary valuations using ordinary circuits; bounded completion searches make this possible.

Rounding five constraints. The rounding input also gives a standalone statement (Theorem 9.7). With a fixed integer coordinate sum, any family of at most five subset inequalities with integer lower bounds that is feasible in $[0, 1]^N$ has a 0/1 solution. Six constraints can fail. The statement applies to arbitrary subsets, beyond the common-center constraints used in the coloring proofs.

Related perspectives. Colorful subgraph theorems of Simonyi and Tardos [ST06] also extract color patterns from topology. Our separation concerns the standard box-complex index bound, within the hierarchy discussed in [DM24]. In fair division, consensus halving for sets of items [GHI+22] permits fractional items and seeks two bundles that each agent values equally. Here the goods remain indivisible, the groups are prescribed, and balance constrains the numbers of goods rather than their values.

Open problems. What is $\chi(G_{n,3})$ for $n \geq 5$, between ten and fourteen? Can the method prove $\chi(G_{n,r}) > 4r - 3$ for $r \geq 4$ and $n \geq 2r - 1$? Can the five-agent balanced EF1 and nine-agent balanced EF2 search problems be solved in polynomial time, and do they admit PPA membership or hardness results? The search equivalence transfers these computational questions without resolving them.

Reading the paper. Section 2 gives the fair-division correspondence. Sections 3 and 4 explain the chromatic proofs; Sections 5 and 6 give the balancing and search results. The detailed reductions, rook coloring argument, and topological proofs follow. Appendices A–C prove the finite integral homology inputs. No theorem depends on computation.

2 Balanced fair division and the chromatic threshold

An agent's rejection family is independent in the allocation graph, and every independent family arises in this way. This correspondence turns a chromatic number into an exact threshold for every prescribed split of the agents.

Table 1 compares the small-agent guarantees. The assumptions differ: Gölz and Yaghoubizade obtain an efficient algorithm and balance under additivity [GY26, Theorem 3.1 and Corollary 3.2], while the revised monotone theorem of Bu, Li, Liu, Song, and Tao concerns two groups of two without asserting cardinality balance [BLL+24, Theorem 3.1].

Work and guarantee	Valuations	Prescribed agents	Balanced
Gölz–Yaghoubizade [GY26], EF1	additive	at most four, every split	yes
[BLL+24], EF1	arbitrary monotone	two groups of two	not asserted
[Dan19] and the chromatic correspondence, EF2	arbitrary monotone	at most five, every split	yes
This paper, EF1	arbitrary monotone	**at most five, every split**	**yes; sharp**
This paper, EF2	arbitrary monotone	**at most nine, every split**	**yes**

Table 1: Group fair-division guarantees. All memberships are fixed in advance. Balance means that the two bundle sizes differ by at most one. The Daneshpajouh row is the consequence of his chromatic bound under the correspondence proved below. The EF1 sharpness is witnessed by six agents on every even number of goods at least four.

The exact chromatic correspondence. We give the proof for general EFc. For EF1, both directions and the 0/1-valued construction are due to [KSV20]; they also noted the extension of the existence argument to EFc.

Let $M = [2n]$ be a set of goods, and fix two nonempty groups of sizes q_1, q_2. Write $q = q_1 + q_2$. Each agent has a normalized monotone valuation $u : 2^M \to \mathbb{R}_{\geq 0}$: $u(\varnothing) = 0$, and $u(S) \leq u(T)$ when $S \subseteq T$. Subtracting $u(\varnothing)$ from every value does not change any envy comparison, so normalization costs nothing. The group sizes and memberships are fixed before the goods are allocated. A balanced allocation is specified by the n-set A given to the first group; the second receives A^c. An agent receiving P accepts the allocation as EFc if

$$u(P) \geq u(Q \setminus T) \quad \text{for some } T \subseteq Q, \quad |T| \leq c,$$

where Q is the other bundle. Otherwise she *rejects* it.

Theorem 2.1 (The balanced threshold)**.** *Let $n > c \geq 1$ and $q_1, q_2 \geq 1$. Every profile of normalized monotone valuations admits a balanced EFc allocation for this split if and only if*

$$q_1 + q_2 < \chi(G_{n,c+1}).$$

If the inequality fails, a counterexample exists with 0/1-valued valuations.

The proof follows the chromatic framework of [KSV20]. We include both directions, since one gives existence and the other gives sharpness.

Lemma 2.2. *A family of balanced allocations is the rejection family of a monotone agent if and only if it is independent in $G_{n,c+1}$. Every independent family is realizable by a normalized 0/1-valued agent in either group.*

Proof. Suppose a first-group agent rejects adjacent allocations A, B. Deleting $A^c \cap B^c$ from the other bundle at A is allowed, since $|A^c \cap B^c| = |A \cap B| \leq c$. Rejection and monotonicity imply

$$u(A) < u(A^c \cap B) \leq u(B).$$

Reversing A and B gives the opposite inequality, a contradiction. For a second-group agent, apply the same argument to complements.

Conversely, let $\mathcal{I}$ be a nonempty independent family. Define

$$u_{\mathcal{I}}(S) = \begin{cases} 0, & S \subseteq B \text{ for some } B \in \mathcal{I}, \\ 1, & \text{otherwise.} \end{cases}$$

This is normalized and monotone. An n-set outside $\mathcal{I}$ has value one, so it is accepted by a first-group agent. For $A \in \mathcal{I}$, suppose that $A^c \setminus T$ has value zero for some $|T| \leq c$. Then $A^c \setminus T \subseteq B$ for some $B \in \mathcal{I}$, and

$$|A \cap B| = |A^c \cap B^c| \leq |T| \leq c.$$

Since $n > c$, the set $A^c \setminus T$ is nonempty, so $B \neq A$. This contradicts independence. Thus exactly the allocations in $\mathcal{I}$ are rejected. For the second group, use the same construction with $\{A^c : A \in \mathcal{I}\}$. The empty family is realized by the identically zero valuation. □

Proof of Theorem 2.1. If every allocation were rejected, color it by the name of one rejecting agent. The lemma makes this a proper q-coloring. Hence $q < \chi(G_{n,c+1})$ guarantees an accepted allocation. Conversely, given a proper coloring with at most q colors, assign one color class to each agent and realize it as her rejection family in the prescribed group. Unused colors give zero valuations. Every allocation is rejected, which proves the converse. □

Consequences and boundary cases. The equality $\chi(G_{n,2}) = 6$ gives balanced EF1 for five agents and, by the tightness construction of [KSV20], a counterexample for every split of six agents on every even number of goods at least four. The bound $\chi(G_{n,3}) \geq 10$ gives balanced EF2 for nine agents when $n \geq 4$. For $n = 3$, $G_{3,3} = K_{20}$, so the same guarantee holds. If $n \leq c$, any balanced allocation is EFc, since the entire comparison bundle may be deleted.

For $2n - 1$ goods, add a dummy good d with $u'(S) = u(S \setminus \{d\})$. Remove d from a balanced EFc allocation on the enlarged set. The bundle sizes now differ by one, and replacing each deletion witness T by $T \setminus \{d\}$ preserves EFc. This proves Theorem 1.1. The dummy-good argument transfers existence, not the even-size counterexamples.

Together with [KSV20, Propositions 3.3 and 3.4, Corollary 3.6], Theorem 1.1 classifies the group-size pairs for unrestricted monotone EF1, except $(5, 1)$, up to exchanging groups. At most five agents always admit EF1; every larger split other than $(5, 1)$ can fail, by those counterexamples and adding zero-valued agents. For $(5, 1)$, balanced EF1 can fail [KSV20, Proposition 4.7], whereas EF1 without the balance requirement is guaranteed for binary additive valuations [KSV20, Theorem 3.1]. The general monotone case is not resolved by these results.

For EF2, the known fourteen-color construction gives counterexamples for every split of fourteen agents on every even number of goods at least eight. Thus the largest universal balanced EF2 agent threshold Q_2 satisfies $9 \leq Q_2 \leq 13$. The upper side is inherited from the known coloring; the new result is the lower side. On exactly eight goods the threshold is 13, since $\chi(G_{4,3}) = 14$.

3 The six-color theorem

We show why five colors cannot suffice for $G_{n,2}$ when $n \geq 3$. There are three steps: a coloring gives a map to a sphere; pentagons represent the homology that this map must preserve; and a count of pairs shows that every pentagon loses a color.

3.1 The coloring map

The *neighborhood complex* $\mathcal{N}(G)$ has as faces the nonempty vertex families with a common neighbor. A simplicial complex records a collection of finite sets, closed under taking subsets; its geometric realization fills each such set by a simplex. For example, three vertices with a common neighbor span a filled triangle in $\mathcal{N}(G)$.

Suppose that $\kappa : V(G) \to [5]$ is a proper coloring. A face of $\mathcal{N}(G)$ cannot use the color of its common neighbor. Thus the coloring defines a simplicial map

$$\kappa_{\mathcal{N}} : \mathcal{N}(G) \longrightarrow \partial\Delta^4 \cong S^3. \tag{3.1}$$

Here Δ^4 is the simplex on the five colors; its boundary consists of all proper subsets of those colors.

We use homology with coefficients in $\mathbb{F}_2$ in the coloring argument. A d-cycle is a sum of d-dimensional simplices whose boundary is zero; two cycles represent the same homology class if their difference is a boundary. A space is t-connected if every map from S^j into it extends over a ball for $0 \leq j \leq t$.

The following fact is the topological constraint on a coloring. We state it for arbitrary d, but use only $d = 3$ in this section. Its proof is given in Section 9.1.

Lemma 3.1 (Nonvanishing)**.** *Let $d \geq 1$, and suppose that $\mathcal{N}(G)$ is $(d-1)$-connected. Every proper coloring with colors in $[d+2]$ induces a nonzero homomorphism*

$$(\kappa_{\mathcal{N}})_* : H_d(\mathcal{N}(G); \mathbb{F}_2) \longrightarrow H_d(\partial\Delta^{d+1}; \mathbb{F}_2) \cong \mathbb{F}_2.$$

For five colors, the induced map must therefore preserve some class in H_3. Our task is to exhibit generators and show that it kills every one. A useful observation is that a cycle supported on five vertices maps to zero whenever two of its vertices have the same color: its image lies in one face of Δ^4, which is contractible.

3.2 Pentagons give the relevant cycles

For k-subsets A, Z of $[N]$, the *Johnson distance* $d_J(A, Z) = k - |A \cap Z|$ is the number of replacements needed to turn A into Z. Let $\mathcal{C}_1(N, k)$ be the complex of families with a common center at Johnson distance at most one. In $G_{n,2}$, complementation turns a common neighbor into a center:

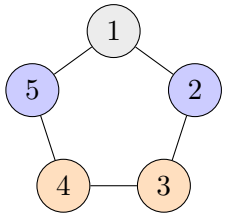
$$\mathcal{N}(G_{n,2}) = \mathcal{C}_1(2n, n). \tag{3.2}$$

Indeed, $|A \cap B| \leq 1$ is equivalent to $d_J(A, [2n] \setminus B) \leq 1$. We call the vertices of this complex *labels* when distinguishing them from centers.

On a five-point set D, arrange a pentagon. Let A_i be the two neighbors of point i, and let Z_i be its two nonneighbors. Then

$$|A_i \cap Z_i| = 0, \qquad |A_i \cap Z_j| = 1 \quad (i \neq j).$$

Thus Z_j is a radius-one center of all four labels other than A_j. Every proper subfamily of $A_1, \ldots, A_5$ is a face of $\mathcal{C}_1(5, 2)$. The five tetrahedra obtained by omitting one label at a time form the boundary of a four-dimensional simplex, hence a three-cycle.



$A_1 = \{2, 5\}$: blue points
$Z_1 = \{3, 4\}$: orange points
Z_j centers the four labels other than A_j

Figure 1: The pentagon provides five labels and five centers. The cycle is the boundary on the five set-labels A_i, not the drawn polygon on the five ground-set points.

To obtain n-sets, choose disjoint $I, D \subseteq [2n]$ with $|I| = n - 2$ and $|D| = 5$, and use the labels $I \cup A_i$. Their centers are $I \cup Z_j$. We also use the cycles obtained by complementing all five n-sets.

Proposition 3.2 (Pentagon generators)**.** *For $n \geq 3$, the complex $\mathcal{N}(G_{n,2})$ is 2-connected, and its $H_3(-; \mathbb{F}_2)$ is spanned by the pentagon cycles just described.*

There are two tasks: compute the homology after allowing fractional centers, then return to integral centers. Replace a set by its 0/1 indicator vector and allow centers to be fractional: $x \in [0, 1]^{2n}$ with $\sum_i x_i = n$. The fractional complex has a geometric model built from coordinate faces of this hypersimplex, obtained by fixing some coordinates to zero or one. The model contains every face of dimension at most three. The first missing faces have five free coordinates, of total weight two or three; their boundaries generate the third homology. Each pentagon cycle projects once onto one of these boundaries, as in Figure 2.

To return to integral centers, an elementary rounding lemma for five labels shows that the relaxation preserves all simplices of dimension at most four, hence the chain groups and boundary maps that compute H_3. Section 9 proves these statements; its short radius-one rounding proof is in Section 9.6. The six-color argument uses no radius-two comparison theorem.

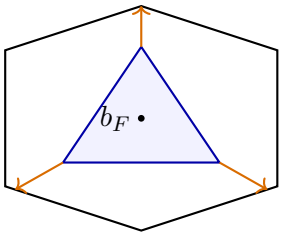


Figure 2: A planar illustration of the degree-one argument. For the pentagon, the inner simplex and coordinate face have dimension four; for the rook graph they have dimension eight. Both contain the same barycenter in their interiors.

3.3 A count of ten pairs

Call a family *rainbow* if its labels have distinct colors. For a pair $Z \subseteq D$, let

$$\mathcal{B}_1(Z) = \{A \in \tbinom{D}{2} : A \cap Z \neq \varnothing\}.$$

We first prove a local statement on just five ground-set points.

Lemma 3.3 (The pentagon loses a color)**.** *Let $\varphi : \binom{D}{2} \to [5]$. If each ball $\mathcal{B}_1(Z)$ misses a color, then the five pentagon labels A_i are not rainbow.*

Proof. Suppose they are rainbow, and name their colors by i. For every pair Z, choose a missing color $\psi(Z)$. The incidence rule forces $\psi(Z_i) = i$. Thus both φ and ψ use all five colors, and

$$\varphi(A) \neq \psi(Z) \quad \text{if } A \cap Z \neq \varnothing.$$

For each color i, let U_i be the union of the pairs given that color by φ. At least one pair has that color, so $|U_i| \geq 2$. A pair colored i by ψ must avoid U_i, so $|U_i| \leq 3$.

Every pair is contained in some U_i, by its φ-color, and in some $D \smallsetminus U_i$, by its ψ-color. If a of the five sets U_i have size three, counting the ten pairs gives

$$10 \leq 5 + 2a, \qquad 10 \leq 15 - 2a.$$

The first inequality requires $a \geq 3$ and the second $a \leq 2$. □

3.4 Finishing the six-color theorem

Suppose that κ is a proper five-coloring of $G_{n,2}$, $n \geq 3$. For a pentagon configuration with fixed core I, put $\varphi(A) = \kappa(I \cup A)$. If $A \in \mathcal{B}_1(Z)$, then

$$|(I \cup A) \cap ([2n] \smallsetminus (I \cup Z))| = 2 - |A \cap Z| \leq 1.$$

These two n-sets are distinct because $I \neq \varnothing$. Thus the entire ball misses the color of $[2n] \smallsetminus (I \cup Z)$, and Lemma 3.3 makes the configuration nonrainbow. The same is true of a complemented configuration: apply the argument to the proper coloring $A \mapsto \kappa(A^c)$, using $|A^c \cap B^c| = |A \cap B|$ for n-sets in $[2n]$.

Every generating three-cycle therefore maps into a proper face of the simplex on the five colors, so $(\kappa_{\mathcal{N}})_*$ is zero on every generator. Proposition 3.2 makes it zero on all of H_3, contradicting Lemma 3.1 with $d = 3$. Hence $\chi(G_{n,2}) \geq 6$.

For the upper bound, fix three points of $[2n]$. Every n-set A or its complement contains at least two of them. Choose one such pair and record the pair and its side. This gives six colors. Sets with the same positive color contain a common pair; for a negative color, their complements do, which gives the same intersection bound. Thus the coloring is proper. When $n = 2$, every two distinct 2-sets of $[4]$ are adjacent, so $G_{2,2} = K_6$. This proves $\chi(G_{n,2}) = 6$ for every $n \geq 2$.

The general coloring criterion. The preceding argument has the following form for an arbitrary graph. It packages the two steps that will also finish the rook proof.

Proposition 3.4. *Let $d \geq 1$. Suppose that $\mathcal{N}(G)$ is $(d-1)$-connected and that cycles z_α generate $H_d(\mathcal{N}(G); \mathbb{F}_2)$. If every proper $(d+2)$-coloring uses at most $d+1$ colors on the support of each z_α, then $\chi(G) \geq d+3$.*

Proof. Each generator maps into a contractible proper face of the simplex on the colors. The induced homomorphism is therefore zero, contradicting Lemma 3.1. □

4 The ten-color lower bound

The coloring map and the final contradiction are the same. There are two new ingredients: nine rook labels replace the five pentagon labels, and identifying the seventh homology requires more than five-constraint rounding. The rook labels define the candidate generators. A comparison between integral and fractional centers establishes their topological role; the local coloring argument then gives the contradiction.

4.1 The rook cycle and the local statement

Number a 3×3 array by $1, \ldots, 9$, row by row, and let D be its set of cells. Let A_i consist of the four other cells in the row or column of i, and put $Z_i = D \smallsetminus (A_i \cup \{i\})$. Then

$$|A_i \cap Z_i| = 0, \qquad |A_i \cap Z_j| = 2 \quad (i \neq j). \tag{4.1}$$

Define $\mathcal{C}_2(N, k)$ using common Johnson centers of radius two. Equation (4.1) says that every proper subfamily of the nine labels is a face of $\mathcal{C}_2(9, 4)$, so its formal boundary is a seven-cycle. For $n \geq 5$, adjoining a fixed $(n-4)$-set I disjoint from D gives a cycle in $\mathcal{N}(G_{n,3}) = \mathcal{C}_2(2n, n)$. Again include the complemented configurations.

1	2	3
4	5	6
7	8	9

$A_1 = \{2, 3, 4, 7\}$: blue cells
$Z_1 = \{5, 6, 8, 9\}$: orange cells
Z_j centers the eight labels other than A_j

Figure 3: The rook configuration plays the role of the pentagon. Its cycle is the boundary on the nine set-labels A_i.

Proposition 4.1 (Rook generators)**.** *For $n \geq 5$, the complex $\mathcal{N}(G_{n,3})$ is 6-connected, and its $H_7(-; \mathbb{F}_2)$ is spanned by the rook cycles just described.*

This is the radius-two case of Proposition 9.1. Its extra comparison step is explained in Section 4.2. The local coloring statement is as follows.

Lemma 4.2 (The rook configuration loses a color)**.** *Let $\varphi : \binom{D}{4} \to [9]$. If each ball $\mathcal{B}_2(Z) = \{A \in \binom{D}{4} : |A \cap Z| \geq 2\}$ misses a color, then the nine labels A_i are not rainbow.*

The local proof, in Section 8, uses two lists of available colors and a short sequence of forced exclusions. Given the lemma, the lower bound follows exactly as before. In a proper nine-coloring, the entire ball around Z misses the color of $[2n]\setminus(I\cup Z)$, because $|(I\cup A)\cap([2n]\setminus(I\cup Z))| = 4-|A\cap Z| \le 2$. The fixed core is nonempty, so this is an edge; complemented configurations are handled by complementing the coloring. Each generating seven-cycle then uses at most eight colors and maps to zero, contradicting Lemma 3.1 with $d = 7$. Therefore $\chi(G_{n,3}) \ge 10$ for $n \ge 5$.

The small case and the upper bound. For $n = 4$, identify the ground set with $\mathbb{F}_2^3$. Its fourteen affine planes have four points each, and two distinct planes meet in zero or two points. They form a clique in $G_{4,3}$. For every $n \ge 4$, the Fano-plane construction gives fourteen colors [BCK19]. Label seven points by the nonzero vectors of $\mathbb{F}_2^3$. The seven triples $\{u, v, u+v\}$ are the Fano lines. Every two-coloring of these points has a monochromatic line. Thus some line lies in A or in A^c; record a chosen line and its side. Two sets with the same color meet in at least three points, using complementation for the negative side. This proves the upper bound and $\chi(G_{4,3}) = 14$.

4.2 The radius-two comparison

The topological input is Proposition 4.1. Its proof requires a comparison between integral and fractional centers: allowing fractional centers must not kill the seventh homology used by the coloring argument.

Replace a k-set by its indicator vector in the hypersimplex

$$\Delta(N,k) = \{x \in [0,1]^N : \textstyle\sum_i x_i = k\}.$$

A common radius-two center satisfies $x(A) \ge k-2$ for every label A, where $x(A) = \sum_{i\in A} x_i$. Requiring $x \in \{0,1\}^N$ gives the integral complex $\mathcal{C}_2(N,k)$. Allowing arbitrary $x \in \Delta(N,k)$ gives a larger complex Y_2, the complex of families with a fractional center. This is a linear relaxation of the common-center condition.

The fractional complex has the coordinate-face model already used for radius one; the rook configurations map onto its seventh-homology generators (Section 9). We must show that the relaxation has not killed any integral homology class in that degree.

The five-constraint rounding theorem (Section 9.7) implies that $\mathcal{C}_2(N,k)$ and Y_2 have the same four-skeleton: their faces with at most five labels agree. At radius one, the shorter lemma in Section 9.6 already identifies the chain groups and boundary maps computing H_3.

For radius two, a seven-cycle has eight-dimensional fillings. Five-label rounding does not control such fillings. We instead prove

$$H_j(\mathcal{C}_2(N,k);\Bbbk) \longrightarrow H_j(Y_2;\Bbbk) \text{ is injective for } 0 \le j \le 7, \qquad \Bbbk = \mathbb{Z}, \mathbb{F}_2. \tag{4.2}$$

Here is the construction. Given a cycle z in the integral complex, we attach compatible center data to obtain a cycle $\widetilde{z}$. Contractibility of the fractional center spaces supplies a filling w whenever z bounds in Y_2. A chain map ϕ returns this filling to the integral complex and satisfies $\phi(\widetilde{z}) = z$. Therefore

$$z = \phi(\partial w) = \partial\phi(w).$$

The equality on the original cycle is essential. Independently rounding the simplices of a filling would not make their boundaries agree.

There are two inputs to this construction. First, a mediator set fills triangles among integral centers. It makes the spaces used for large label families simply connected; small families use the

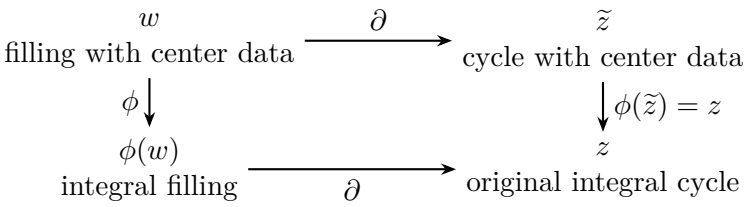


Figure 4: Returning a fractional filling to the integral complex. The chain map is constructed through degree eight and agrees exactly with the prescribed map on the lifted cycle.

contractible fractional center spaces and five-label rounding. Second, the return map is extended one simplex at a time inside local complexes of integral centers. The required vanishings reduce to

$$\widetilde{H}_j(\mathcal{C}_2(9,3);\mathbb{Z}) = \widetilde{H}_j(\mathcal{C}_2(8,4);\mathbb{Z}) = 0 \qquad (0 \le j \le 7). \tag{4.3}$$

These fixed complexes have 84 and 70 labels. The coordinate-face analysis in Section 9.3 explains why they suffice for every N, k; Section 11 gives the comparison. Appendices B and C prove (4.3) over the integers, as needed to deduce connectivity. The final coloring argument uses only $\mathbb{F}_2$.

5 Balancing the full allocation graph

The graph on all allocations has the same chromatic number as its balanced layers. This permits discrepancy bounds proved without a balance requirement to yield balanced fair allocations.

For m goods, let $\Gamma_{c,c}(m)$ have all subsets of $[m]$ as vertices, with distinct A, B adjacent when

$$|A \cap B| \le c, \qquad |A^c \cap B^c| \le c.$$

Complements are taken in the current ground set. For $c = 1$, this is the graph used in the revised proof of [BLL+24, Definition 3.3]. The *central layers* consist of sets of sizes $\lfloor m/2 \rfloor$ and $\lceil m/2 \rceil$. A graph retraction onto these layers preserves edges and fixes every central vertex.

Theorem 5.1 (Exact balancing)**.** *For $c \ge 1$ and $m \ge 2c+1$, there is a complement-preserving graph retraction from $\Gamma_{c,c}(m)$ onto its central layers, computable in $O(m)$ time. These layers induce a graph isomorphic to $G_{\lceil m/2 \rceil, c+1}$. Consequently,*

$$\chi(\Gamma_{c,c}(m)) = \chi(G_{\lceil m/2 \rceil, c+1}).$$

On $[2n]$, define $\beta(X)$ by adding the smallest missing elements if $|X| < n$, or deleting the smallest present elements if $|X| > n$, until its size is n. This fixes every n-set and preserves complements: the additions for X are precisely the deletions for X^c.

Lemma 5.2 (The balancing scan is nonexpanding)**.** *For $\delta(X,Y) = \max\{|X \smallsetminus Y|, |Y \smallsetminus X|\}$, one has $\delta(\beta(X), \beta(Y)) \le \delta(X,Y)$.*

Proof. Implement β by one scan through the coordinates, changing a coordinate only if this moves the set toward size n. Apply each scan step to both sets. If $|X \smallsetminus Y|$ increases, it increases by one. This can happen only when X adds that coordinate, Y deletes it, or both do so. The three cases imply, respectively, $|X| < n \le |Y|$, $|X| \le n < |Y|$, or $|X| < n < |Y|$ before the step. Thus

$$|X \smallsetminus Y| + 1 \le |Y \smallsetminus X|,$$

because $|Y \smallsetminus X| - |X \smallsetminus Y| = |Y| - |X|$. The increased count does not exceed the old maximum. Interchanging X, Y treats the other count, so no scan step increases δ. □

Proof of Theorem 5.1. First let $m = 2n$. For n-sets, $|A^c \cap B^c| = |A \cap B|$, so the central layer induces $G_{n,c+1}$. Adjacency in the full graph means $\delta(A, B^c) \le c$. Lemma 5.2 and complement preservation give

$$|\beta(A) \cap \beta(B)| = \delta(\beta(A), \beta(B)^c) = \delta(\beta(A), \beta(B^c)) \le c.$$

The images are distinct because $n > c$. Hence β is the required retraction.

For $m = 2n - 1$, adjoin a coordinate ∞ and put

$$\theta(A) = \begin{cases} A \cup \{\infty\}, & |A| < n, \\ A, & |A| \ge n. \end{cases}$$

This preserves complements. To check edges, write $a = |A \cap B|$ and $b = |A^c \cap B^c|$. Then $a - b = |A| + |B| - (2n - 1)$. If both sets have size less than n, then $a < b$ and their new intersection counts are $a + 1, b$; if both have size at least n, then $b < a$ and the counts are $a, b + 1$. Otherwise the counts are unchanged. Thus θ preserves edges.

On the central layers, θ is a bijection onto the n-sets of $[2n]$. The same identity shows that the image intersection has size $\max\{a, b\}$, so this bijection is a graph isomorphism onto $G_{n,c+1}$. Compose θ with the even balancing map and then delete ∞. This is a complement-preserving graph map into the central layers and fixes every central vertex. Size counts and one scan take $O(m)$ time. Finally, the retraction and the inclusion of the central layers give the two chromatic inequalities. □

The retraction transfers chromatic existence guarantees. It need not preserve the fairness of an allocation for fixed valuations, so it does not give a procedure for rebalancing an arbitrary fair allocation.

5.1 General removal allowances

Hollender, Manurangsi, Meka, and Suksompong already guarantee EFc with $c = O(\sqrt{q \log q})$ for q agents with monotone valuations, without requiring balance [HMMS26, Corollary 11]. Their nonlinear discrepancy theorem, together with Theorem 5.1, gives the same order of allowance with balance. Independently and concurrently, Dupré la Tour and Fujii [DF25, Theorem 1] proved the discrepancy estimate, also for any prime-power number of parts.

A set function is 1-*Lipschitz* if adding or removing one element changes its value by at most one. The two-part case of [HMMS26, Theorem 8] says that, for any q such functions on $2^{[m]}$, some $S \subseteq [m]$ satisfies

$$|f_i(S) - f_i(S^c)| < D(q) := \sqrt{2q(1 + \ln(2q))} \qquad (i \in [q]).$$

Corollary 5.3. *If $q \ge 2$, $c \ge 1$, and $n > c$ are integers with $D(q) \le c + 1$, then $\chi(G_{n,c+1}) > q$. Consequently, for every prescribed split of $q \ge 2$ agents with monotone valuations and any number of goods, balanced EFc is guaranteed with $c = \lceil D(q) \rceil - 1 = O(\sqrt{q \log q})$.*

Proof. Suppose that $G_{n,c+1}$ has a proper q-coloring. Composing it with the retraction gives a proper coloring of $\Gamma_{c,c}(2n)$. For each nonempty color class $\mathcal{I}_i$, use its distance function $f_i(S) = \min_{T \in \mathcal{I}_i} |S \triangle T|$. These functions are 1-Lipschitz; repeat them to obtain q if needed. Choose S from the discrepancy theorem and let i be its color. Since $f_i(S) = 0$, there is $T \in \mathcal{I}_i$ with $|S^c \triangle T| < D(q) \le c + 1$. Integrality gives

$$|S \cap T| + |S^c \cap T^c| = |S^c \triangle T| \le c.$$

Thus S, T are adjacent and have the same color. They are distinct, since $S = T$ would make the displayed distance $2n > c$. This contradiction proves the chromatic bound. Theorem 2.1 now gives

balanced EFc when $n > c$. The small-bundle and dummy-good arguments in Section 2 cover every remaining number of goods. □

6 Equivalent search problems

The existence correspondence also relates the computational problems when the valuations and coloring are specified by circuits. The allowance c is fixed throughout this section. Let $n > c$, fix positive group sizes q_1, q_2, and put $q = q_1 + q_2$ and $M = [2n]$. Agents $1, \dots, q_1$ are in the first group. Write $P_i(X) = X$ for these agents and $P_i(X) = X^c$ for the others; thus $P_i(X)$ is agent i's bundle at allocation (X, X^c).

Definition 6.1 (The circuit search problems)**.** In *balanced Kneser search*, the input is a Boolean circuit computing a color $\varphi(X) \in [q]$ for each n-set X. Outputs outside $[q]$ are interpreted as color one. A solution is two n-sets X, Y with $\varphi(X) = \varphi(Y)$ and $|X \cap Y| \leq c$.

In *balanced EFc search*, the input is one rational-valued circuit $u_i : 2^M \to \mathbb{Q}$ per agent, with no monotonicity promise. A solution is either

(i) an n-set X and, for each agent i, a set $D_i \subseteq P_i(X)^c$ with $|D_i| \leq c$ and $u_i(P_i(X)) \geq u_i(P_i(X)^c \setminus D_i)$; or

(ii) an agent i and sets $S \subsetneq T$ with $u_i(S) > u_i(T)$.

Let L include the circuit descriptions, input widths, and output widths; Section 7 specifies the encoding. Solutions have polynomial length and can be checked in polynomial time. The violation alternative, as in [Hav24, Remark 6.6], makes totality independent of a monotonicity promise.

Theorem 6.2 (Search equivalence)**.** *For fixed $c \geq 1$ and positive q_1, q_2, suppose that $n > c$ and $q = q_1 + q_2 < \chi(G_{n,c+1})$. The two problems in Definition 6.1 are equivalent under deterministic polynomial-time many-one search reductions. Both reductions preserve n, c, q, take time $L^{O(1)}n^{O(c)}$, and produce circuits of that size. The reduction from Kneser search produces normalized monotone $\{0,1\}$-valued valuations.*

Both problems are total under the stated chromatic inequality.

The forward reduction colors an allocation by a rejecting agent. A monochromatic edge then supplies either a fair allocation or a monotonicity violation. For the reverse reduction, the downward closure of a color class suggests a binary valuation, but evaluating that closure could require exponentially many color queries. Assigning zero to every bundle smaller than $n - c$ avoids this search: every other relevant bundle has at most c missing elements in an n-set. Section 7 gives both reductions and their decoders in full.

Haviv's reverse reduction from classical Kneser search to Agreeable-Set uses subset queries asking whether a set contains a vertex of a specified color [Hav24, Theorem 1.5]. Here the bounded completion search implements the reverse reduction with ordinary circuits.

In particular, the five-agent balanced EF1 and nine-agent balanced EF2 problems, for $n \geq 2$ and $n \geq 3$ respectively, lie in TFNP: solutions always exist and have polynomially bounded, polynomially verifiable certificates. Polynomial-time algorithms and membership or hardness in PPA, the class based on undirected parity, remain open.

7 The search reductions

We now prove Theorem 6.2. The forward construction tests at most c deletions from a bundle. The reverse construction tests at most c additions to one. These bounded searches make both reductions polynomial for fixed c. We use the notation of Section 6.

Subsets are encoded by indicator vectors. A valuation circuit outputs a sign bit s and unsigned integers a, b, decoded as $(-1)^s a/(b+1)$; hence every output string has a value. Comparisons use exact integer arithmetic. A many-one search reduction constructs one target instance and converts every valid target solution into a source solution. On monotone inputs only allocations can be returned; subtracting $u_i(\varnothing)$ normalizes such inputs without changing fairness. Both problems are total under the stated chromatic inequality: Theorem 2.1 applies to monotone inputs, and a nonmonotone input has a violation certificate. The polynomial bound requires fixed c; it is not a bound of the form $f(c)\,\mathrm{poly}(L, n)$.

7.1 From valuations to a monochromatic edge

Given the valuation circuits, color a balanced allocation X by its first rejecting agent, or by color one if all agents accept it. For each agent, acceptance can be tested by enumerating all $\sum_{j=0}^{c} \binom{n}{j} = O(n^c)$ deletion sets from the other bundle. Composing these tests with the valuation circuits gives a coloring circuit of size $L^{O(1)} n^{O(c)}$, without testing monotonicity.

Let X, Y be any monochromatic edge returned by the target problem. Test both allocations. If either is fair, return it with the deletion witnesses found by the tests. Otherwise their common color identifies an agent i who rejects both, including when this color is one. Put $P_X = P_i(X)$, $P_Y = P_i(Y)$ and $Q_X = P_X^c$, $Q_Y = P_Y^c$. Interchange X, Y if necessary so that $u_i(P_X) \geq u_i(P_Y)$. Balance and adjacency give

$$|Q_X \cap Q_Y| = |X \cap Y| \leq c.$$

Rejection at X applies in particular to deleting $Q_X \cap Q_Y$ from Q_X. The remaining set $S = Q_X \cap P_Y$ satisfies

$$S \subseteq P_Y, \qquad u_i(S) > u_i(P_X) \geq u_i(P_Y).$$

The values differ, so $S \subsetneq P_Y$. Thus (i, S, P_Y) is a valid monotonicity-violation certificate. Every target solution has been decoded in polynomial time, proving the forward reduction.

7.2 From a coloring to binary valuations

Let φ be the input coloring. Assign color i to agent i in the prescribed split, and define $u_i : 2^M \to \{0, 1\}$ by

$$u_i(S) = 0 \quad \Longleftrightarrow \quad \begin{cases} |S| < n - c, \quad \text{or} \\ |S| \leq n \text{ and some } Z \in \binom{M}{n} \text{ contains } S \text{ and satisfies } \varphi(P_i(Z)) = i. \end{cases} \tag{7.1}$$

Otherwise $u_i(S) = 1$. The zero sets are downward closed: a subset of a small zero set is still small, and a subset of a set admitting the displayed completion admits that same completion. Thus u_i is monotone for every coloring, even when its color classes are not independent. It is normalized because $n > c$.

For $|S| < n - c$ or $|S| > n$, the value is immediate. Otherwise enumerate all sets of at most c additional goods, retain those whose union with S has size n, and evaluate φ on the corresponding $P_i(Z)$. There are $O(n^c)$ candidates. This enumeration can be built into an ordinary Boolean circuit, using one copy of the coloring circuit for each candidate and taking the disjunction of the tests.

The circuit size and construction time are $L^{O(1)}n^{O(c)}$. The small-bundle clause is what makes the test bounded; it does not affect any comparison of a balanced bundle with the other bundle after at most c deletions.

Every valid solution for these monotone valuations is a fair allocation with deletion witnesses. Let X be its first-group bundle and take $i = \varphi(X)$. Agent i receives $P = P_i(X)$, and $u_i(P) = 0$: in (7.1), choose $Z = P$ and use $P_i(P_i(X)) = X$. Her deletion witness $D_i \subseteq P^c$ gives $u_i(P^c \setminus D_i) = 0$. Since $|P^c \setminus D_i| \geq n - c$, the small-bundle clause cannot explain this value. Enumerating completions therefore finds an n-set $Z \supseteq P^c \setminus D_i$ with $\varphi(P_i(Z)) = i$. At most $|D_i|$ elements of Z lie in P, so $|P \cap Z| \leq c$.

Output $Y = P_i(Z)$. Complementation preserves intersection sizes of two n-sets in $[2n]$, and hence

$$|X \cap Y| = |P \cap Z| \leq c, \qquad \varphi(Y) = i = \varphi(X).$$

The vertices are distinct because $n > c$. The decoder uses $O(n^c)$ color evaluations, completing the proof of Theorem 6.2.

For EF1, Theorem 1.2 gives totality for $2 \leq q \leq 5$ and $n \geq 2$. For EF2 it gives totality for $2 \leq q \leq 9$ and $n \geq 4$; the remaining case $n = 3$ follows from $G_{3,3} = K_{20}$. These equivalences hold for every positive split and also for binary-valued input circuits, with the same violation alternative. The forward reduction accepts such inputs, and the reverse reduction already produces them.

8 The local rook coloring obstruction

We prove Lemma 4.2. If the nine rook labels had distinct colors, each four-set would have a short list of available colors on each side of the incidence relation. An attachment restriction reduces these lists, after which four forced values give a contradiction.

8.1 Two lists

We use strings for small sets: 1234 means $\{1, 2, 3, 4\}$. The count of pairs from the pentagon proof is replaced by restrictions on the available colors.

Let φ be as in Lemma 4.2, and suppose that the values $\varphi(A_1), \ldots, \varphi(A_9)$ are distinct. Renaming the values, we may assume $\varphi(A_i) = i$. For every 4-set $Z \subseteq D$ choose a value $\psi(Z)$ not occurring on $\mathcal{B}_2(Z)$. Then

$$\varphi(S) \neq \psi(T) \qquad \text{whenever } |S \cap T| \geq 2. \tag{8.1}$$

By the incidence rule (4.1), $\mathcal{B}_2(Z_i)$ contains every A_j with $j \neq i$, hence every value other than i; so $\psi(Z_i) = i$. Applying (8.1) with $T = Z_i$, and with $S = A_i$, gives

$$\varphi(S) \in L(S) = \{i : \ |S \cap Z_i| \leq 1\}, \qquad \psi(T) \in L^*(T) = \{i : \ |T \cap A_i| \leq 1\}. \tag{8.2}$$

Lists are easily computed. For a cell i in row a and column b, let r_a and c_b be the numbers of elements of S in row a and in column b, and $e_i = 1$ if $i \in S$, $e_i = 0$ otherwise. Then $|S \cap A_i| = r_a + c_b - 2e_i$ and $|S \cap Z_i| = 4 - r_a - c_b + e_i$, so that

$$i \in L(S) \iff r_a + c_b - e_i \geq 3, \qquad i \in L^*(T) \iff r_a + c_b - 2e_i \leq 1 \ \text{ (counts taken in } T). \tag{8.3}$$

For example, $L(1234) = 123$: in 1234 the row counts are $3, 1, 0$ and the column counts $2, 1, 1$.

We write $F_S = \varphi(S)$ and $G_T = \psi(T)$, and denote sets by strings of digits; thus $F_{1234} = \varphi(\{1, 2, 3, 4\})$. A *solution* is a pair of maps $F, G\colon \binom{D}{4} \to [9]$ satisfying (8.1) with $F_{A_i} = i$ for every i; as shown above, it also satisfies $G_{Z_i} = i$ and (8.2). We shall show that there is no solution. All lists and intersections used below are read off from (8.3).

8.2 Symmetries

Two symmetries avoid repeating the case analysis. Grid symmetries reduce the attachment restriction to one representative; a duality then transfers the restriction from F to G.

Automorphisms. An automorphism g of the rook graph permutes the cells and satisfies $g(A_i) = A_{g(i)}$ and $g(Z_i) = Z_{g(i)}$. If (F, G) is a solution, so is (F', G') with $F'_S = g(F_{g^{-1}S})$ and $G'_T = g(G_{g^{-1}T})$. The automorphisms include the permutations of rows, the permutations of columns, and transposition. They act transitively on the triples (Λ, e, i) in which Λ is a line (a row or a column), e is a cell outside Λ, and i is the cell of Λ lying in a common line with e.

Duality. Write the cell in row a and column b as (a, b), and put $\pi(a, b) = (a + b,\, a - b)$ modulo 3. Then π is a bijection which interchanges adjacency and nonadjacency of distinct cells: if two cells differ by (α, β), their images differ by $(\alpha + \beta, \alpha - \beta)$, and modulo 3 exactly one of α, β vanishes if and only if neither of $\alpha + \beta, \alpha - \beta$ does. Consequently $\pi(A_i) = Z_{\pi(i)}$ and $\pi(Z_i) = A_{\pi(i)}$. In the numbering of D,

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 1 & 6 & 8 & 5 & 7 & 3 & 9 & 2 & 4 \end{pmatrix}.$$

If (F, G) is a solution, so is $(F^\vee, G^\vee)$, where

$$F^\vee_S = \pi^{-1}(G_{\pi(S)}), \qquad G^\vee_T = \pi^{-1}(F_{\pi(T)}).$$

Indeed (8.1) is preserved because $|\pi S \cap \pi T| = |S \cap T|$, and $F^\vee_{A_i} = \pi^{-1}(G_{Z_{\pi(i)}}) = i$.

8.3 The attachment restriction

Lemma 8.1. *For every solution, $F_{\Lambda \cup \{e\}} \neq i$ whenever Λ is a line, e a cell outside Λ, and i the cell of Λ lying in a common line with e. In particular $F_{1234} \neq 1$ and $F_{1247} \neq 1$.*

Proof. By transitivity it suffices to prove $F_{1234} \neq 1$ for every solution. (The second example is the image of the first under transposition.) Suppose that $F_{1234} = 1$. One has $F_{5689} \in L(5689) = 5689$. The column exchange $\tau = (2\,3)(5\,6)(8\,9)$ fixes the cell 1 and the sets 1234 and 5689; replacing the solution by its image under τ if necessary, we may assume $F_{5689} \in \{5, 8\}$. In each case the table below lists, in order, variables whose values are forced. A removed value v is followed, in parentheses, by a variable already known to take the value v on a set meeting the current set in at least two cells; by (8.1) the current variable cannot take the value v. The symbol $\varnothing$ marks a contradiction.

variable	list	removed values	value
Case $F_{1234} = 1$, $F_{5689} = 5$			
G_{1259}	159	1 (F_{1234}), 5 (F_{5689})	9
G_{1369}	158	1 (F_{1234}), 5 (F_{5689})	8
F_{2679}	389	8 (G_{1369}), 9 (G_{1259})	3
F_{1789}	789	8 (G_{1369}), 9 (G_{1259})	7
G_{3579}	357	3 (F_{2679}), 5 (F_{5689}), 7 (F_{1789})	$\varnothing$
Case $F_{1234} = 1$, $F_{5689} = 8$			
G_{1456}	189	1 (F_{1234}), 8 (F_{5689})	9
G_{1369}	158	1 (F_{1234}), 8 (F_{5689})	5
G_{1368}	168	1 (F_{1234}), 8 (F_{5689})	6
F_{3468}	569	5 (G_{1369}), 6 (G_{1368}), 9 (G_{1456})	$\varnothing$

For instance, the first line uses $1259 \cap 1234 = 12$ and $1259 \cap 5689 = 59$. Both cases are impossible. □

Corollary 8.2. *For every solution, $G_{1249} \neq 9$ and $G_{1267} \neq 6$.*

Proof. Apply Lemma 8.1 to the dual solution $(F^\vee, G^\vee)$: with $\Lambda = 789$, $e = 1$, $i = 7$, and with $\Lambda = 258$, $e = 1$, $i = 2$, it gives $F^\vee_{1789} \neq 7$ and $F^\vee_{1258} \neq 2$. Since $\pi(1789) = 1249$, $\pi(7) = 9$, $\pi(1258) = 1267$ and $\pi(2) = 6$, these read $G_{1249} \neq 9$ and $G_{1267} \neq 6$. □

8.4 Completion of the local coloring argument

Let (F, G) be a solution. By Lemma 8.1, $F_{1234} \in L(1234) \setminus \{1\} = \{2, 3\}$. The exchange τ of the preceding proof fixes 1234 and interchanges the values 2 and 3, and Lemma 8.1 and Corollary 8.2 hold for every solution; so we may assume $F_{1234} = 2$. Then successively

$$G_{1249} \in L^*(1249) \setminus \{9, 2\} = 249 \setminus \{9, 2\} = \{4\},$$
$$F_{1247} \in L(1247) \setminus \{1, 4\} = 147 \setminus \{1, 4\} = \{7\},$$
$$G_{1267} \in L^*(1267) \setminus \{6, 7\} = 267 \setminus \{6, 7\} = \{2\}.$$

In each line the first removed value comes from Corollary 8.2 or Lemma 8.1, and the second from the preceding line together with (8.1), using $1249 \cap 1234 = 124$, $1247 \cap 1249 = 124$ and $1267 \cap 1247 = 127$. But $1267 \cap 1234 = 12$, so (8.1) forbids $G_{1267} = F_{1234} = 2$. Hence there is no solution, and the values $\varphi(A_1), \ldots, \varphi(A_9)$ are not distinct.

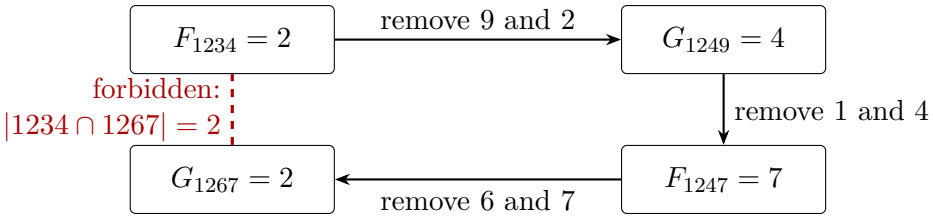


Figure 5: The last four forced values in the rook argument. Each solid arrow uses an already proved restriction and the preceding value. The dashed edge contradicts the rule that intersecting opposite-side sets must receive different colors.

9 The common topological and geometric tools

We prove the statements used in the common coloring argument. The nonvanishing lemma applies to arbitrary graphs, and the geometric construction applies to every integral radius. Radius one has a short rounding argument; radius two uses the general rounding theorem and an additional comparison. A reader following the six-color proof may set $c = 1$ throughout and skip the indicated radius-two material.

9.1 Why the coloring map is nonzero

Proof of Lemma 3.1. Let $B(G)$ be the geometric realization of $\mathrm{Hom}(K_2, G)$. Concretely, its points are pairs (λ, μ) of probability distributions on $V(G)$ such that every vertex in the positive support of λ is adjacent to every vertex in the positive support of μ. Each complete bipartite subgraph gives a cell, the product of the two simplices of weights on its sides. This is a finite free $\mathbb{Z}_2$-CW complex,

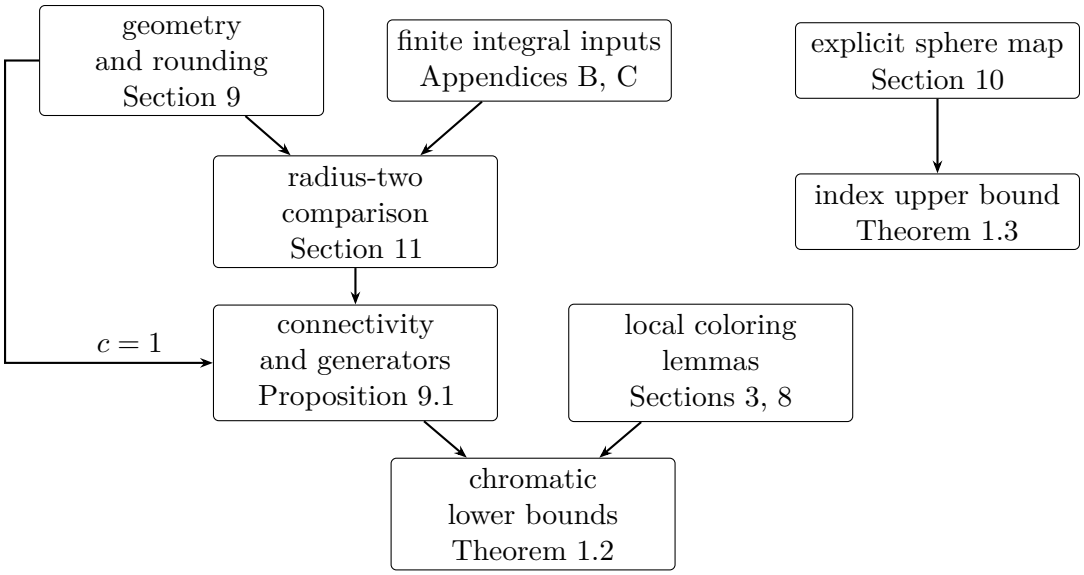


Figure 6: Proof dependencies. Radius one uses the common geometry and rounding. Radius two also uses the comparison and finite integral inputs. The sphere map proves the index upper bound independently; connectivity supplies the matching lower bounds for $r = 2, 3$.

with the involution swapping the two distributions. Babson and Kozlov prove that $B(G) \simeq |\mathcal{N}(G)|$ [BK06, Proposition 4.2]; it is therefore $(d-1)$-connected.

There is consequently an equivariant map $a : S^d \to B(G)$. Here is the construction behind this assertion. Choose a point of $B(G)$ and send an antipodal pair to that point and its image under the involution. Join the two images by a path, and use the involution to define the image of the opposite semicircle. Continue by extending over one hemisphere and defining the other by symmetry. The extension at each stage is possible because $B(G)$ is $(d-1)$-connected. This is the argument in the proof of [Mat03, Proposition 5.3.2(iv), pp. 97–98].

Let $q = d + 2$, and sum the weights of equally colored vertices to obtain $x_i = \sum_{\kappa(v)=i} \lambda_v$ and $y_i = \sum_{\kappa(v)=i} \mu_v$ for $i \in [q]$. Properness makes the supports of x and y disjoint. Thus

$$h(\lambda, \mu) = \frac{x - y}{\|x - y\|} \quad \text{lies on the unit sphere in } \{z \in \mathbb{R}^q : \textstyle\sum_i z_i = 0\}.$$

This is an equivariant map to S^d. The composite $h \circ a$ is an antipodal self-map of S^d, hence has odd degree and is nonzero on $H_d(-; \mathbb{F}_2)$ [Mat03, Proposition 2.4.1(iii) and the following notes, pp. 43–46].

To connect this conclusion to the coloring map, let $p(\lambda, \mu) = \lambda \in |\mathcal{N}(G)|$. This is well-defined because any vertex in the support of μ is a common neighbor of the support of λ. Its colored image is x on the simplex boundary. Put $b = (1/q, \ldots, 1/q)$ and let $r(x) = (x - b)/\|x - b\|$ identify this boundary with the sphere. The segment from $x - b$ to $x - y$ never passes through zero: any coordinate with $y_i > 0$ has $x_i = 0$ and remains strictly negative. Normalizing the segment gives $h \simeq r \circ \kappa_{\mathcal{N}} \circ p$. If $(\kappa_{\mathcal{N}})_*$ were zero, $(h \circ a)_*$ would therefore be zero, a contradiction. □

9.2 One notation for the two configurations

The pentagon and rook arguments have already been stated separately. For their geometric proofs, use a radius c and write $\mathcal{C}_c(N, k)$ for the complex of families of k-sets with a common Johnson

center at distance at most c. For $n > c$,

$$\mathcal{N}(G_{n,c+1}) = \mathcal{C}_c(2n, n).$$

This is the same complementation identity used in the two lower bounds.

For $c = 1, 2$, put $q = 4c + 1$. On a q-point set D, use the pentagon or rook graph, and write A_i for the neighbors of i and $Z_i = D \setminus (A_i \cup \{i\})$. In both cases

$$|A_i| = |Z_i| = 2c, \qquad |A_i \cap Z_i| = 0, \qquad |A_i \cap Z_j| = c \quad (i \neq j).$$

Choose I disjoint from D. If $|I| = k - 2c$, use the labels $I \cup A_i$; if $|I| = k - (2c+1)$, use $I \cup (D \setminus A_i)$. Their centers are obtained in the same way from Z_j. Denote either boundary cycle by $z_{I,D}$, using only choices for which the specified sizes are possible.

Proposition 9.1 (Connectivity and generators)**.** *For $c \in \{1, 2\}$ and $1 \leq k < N$, the complex $\mathcal{C}_c(N, k)$ is $(4c - 2)$-connected. Its homology group $H_{4c-1}(\mathcal{C}_c(N, k); \mathbb{F}_2)$ is spanned by the classes $[z_{I,D}]$.*

Taking $N = 2n$ and $k = n$ gives Propositions 3.2 and 4.1. Indeed, complementation of an n-set of the first type gives one of the second type, and conversely. For $c = 1$, every ingredient in the proof is in this section. For $c = 2$, only the final step uses the comparison theorem from Section 11.

9.3 The hypersimplex and its caps

We now work with arbitrary $1 \leq k < N$ and integer radius $c \geq 1$. For $S \subseteq [N]$ write $x(S) = \sum_{i \in S} x_i$, and let $\mathbf{1}_S$ denote its indicator vector. In the hypersimplex $\Delta(N, k)$ define the cap

$$P_{c,Z} = \{x \in \Delta(N, k) : x(Z) \geq k - c\}, \qquad U_c = \bigcup_{|Z|=k} P_{c,Z}.$$

The integral vertices of $P_{c,Z}$ are precisely the labels at distance at most c from Z. Sending A to $\mathbf{1}_A$ and extending affinely over each simplex defines

$$f_c : |\mathcal{C}_c(N, k)| \longrightarrow U_c. \tag{9.1}$$

It is well-defined because all labels of a face have a common center Z, and their indicator vectors lie in the convex cap $P_{c,Z}$.

A coordinate face F is specified by disjoint sets $I, O \subseteq [N]$: coordinates in I are fixed to one, those in O to zero, and the remaining set D is free. Put $m = |D|$ and $\ell = k - |I|$. Then F is a copy of $\Delta(m, \ell)$. It is a point if $\ell = 0$ or m; otherwise it has dimension $m - 1$. Define

$$\rho(F) = \frac{\ell(m - \ell)}{m}, \qquad \rho(F) = 0 \text{ for a point.}$$

Call F *retained* when $\rho(F) \leq c$, and let Q_c be the union of retained faces. For a point x, let $F(x)$ be its smallest coordinate face. Its free coordinates are exactly those with $0 < x_i < 1$.

Lemma 9.2 (The face model)**.** *(a) A face is contained in U_c if and only if it is retained. Every face of a retained face is retained.*

(b) There is a strong deformation retraction $U_c \to Q_c$ which preserves each cap at every time. On an omitted face whose proper faces are retained, the final retraction is radial projection from its barycenter to its boundary.

(c) Q_c is $(4c-2)$-connected.

Proof. On a coordinate face, the function $\max_{|Z|=k} x(Z)$ is convex and invariant under permutations of the free coordinates. Averaging over these permutations shows that its minimum occurs at the barycenter b_F. Its value there is $k - \rho(F)$. This proves the first assertion of (a). Fixing a further coordinate to zero or one decreases ρ, so retained faces form a subcomplex of the face structure.

If F is omitted, its barycenter lies outside every cap. Push points of $F \cap U_c$ radially away from b_F until they reach ∂F. This motion preserves any cap that contains the starting point: the linear score $x(Z)$ increases along that ray, since its value at b_F is below $k - c$. Process omitted faces in decreasing dimension. Each motion fixes the boundary of its face, so the motions fit together and fix Q_c. This proves (b).

If $m \leq 4c$, then $\rho(F) \leq m/4 \leq c$. Hence Q_c contains every face of dimension at most $4c-1$. The contractible polytope $\Delta(N,k)$ is obtained from Q_c by attaching cells of dimension at least $4c$. Cellular approximation and the relative homotopy sequence give $\pi_j(Q_c) = 0$ for $j \leq 4c-2$ [Hat02, Chapter 4]. □

For $c = 1, 2$, the first omitted faces have $4c+1$ free coordinates. Substitution into the inequality $\ell(m-\ell) > cm$ gives

Radius	Cycle dimension	First omitted faces	Configuration
1	3	$\Delta(5,2),\ \Delta(5,3)$	pentagon
2	7	$\Delta(9,4),\ \Delta(9,5)$	rook graph

Lemma 9.3 (Boundary generators)**.** *Let $c = 1$ or 2, and put $d = 4c-1$. Over $\mathbb{Z}$ or $\mathbb{F}_2$, the group $H_d(Q_c)$ is generated by the boundary classes of the omitted faces listed in the table.*

Proof. The relative cellular chain complex of $(\Delta(N,k), Q_c)$ has no cells in dimensions at most d. Thus its $(d+1)$st homology is a quotient of the free group on omitted $(d+1)$-faces. The boundary homomorphism of the pair identifies this group with $H_d(Q_c)$ and sends each face to its boundary class. Higher-dimensional cells may impose relations among these classes; they do not change the generating set. □

9.4 The fractional complex and the affine map

The nerve Y_c of the caps has as faces the nonempty families σ for which $P_{c,\sigma} := \bigcap_{A \in \sigma} P_{c,A}$ is nonempty. Its points are fractional centers. An integral center supplies a fractional one, giving an inclusion $i_c : \mathcal{C}_c(N,k) \hookrightarrow Y_c$. Because cap intersections are convex, the nerve theorem gives $|Y_c| \simeq U_c \simeq Q_c$.

The inclusion and the affine map have the same homological information, up to the natural identifications from the two covers. We record this fact to avoid confusing the two maps later.

Lemma 9.4 (Nerve comparison)**.** *For $\Bbbk = \mathbb{Z}$ or $\mathbb{F}_2$, there are isomorphisms*

$$\alpha : H_*(\mathcal{C}_c(N,k);\Bbbk) \longrightarrow H_*(\mathcal{C}_c(N,k);\Bbbk), \qquad \beta : H_*(Y_c;\Bbbk) \longrightarrow H_*(U_c;\Bbbk)$$

such that $(f_c)_* = \beta \circ (i_c)_* \circ \alpha$.

Proof. Let L_Z be the full simplex on the labels within distance c of Z. These simplices cover $\mathcal{C}_c(N,k)$. Their intersections are simplices; by symmetry of Johnson distance, the intersection indexed by a family ς is nonempty exactly when $\varsigma \in \mathcal{C}_c(N,k)$. Thus the nerve of this cover is another copy of $\mathcal{C}_c(N,k)$. The cap cover of U_c has nerve Y_c, and $f_c(L_Z) \subseteq P_{c,Z}$.

Write $C = \mathcal{C}_c(N,k)$. The auxiliary spaces for the two covers are

$$X = \bigcup_{\varsigma \in C} |\varsigma| \times \Big| \bigcap_{Z \in \varsigma} L_Z \Big|, \qquad X' = \bigcup_{\varsigma \in Y_c} |\varsigma| \times P_{c,\varsigma}.$$

All four coordinate projections are homotopy equivalences by the nerve theorem for these finite good covers [Bjo95], using compatible triangulations and [Hat02, Proposition 4G.1 and Section 4.G, Exercise 4, pp. 458–460]. The map $(s,t) \mapsto (s, f_c(t))$ from X to X' commutes with the projections. Using the first projections identifies its map on homology with $(i_c)_*$; using the second identifies it with $(f_c)_*$. This gives the stated formula. □

9.5 The degree of a configuration

We return to the cycles $z_{I,D}$ of Section 9.2. Their integral vertices lie on a coordinate face of the appropriate type in the table above. The following argument treats both small graphs at once.

Lemma 9.5 (Degree-one representatives)**.** *For $c = 1, 2$, the affine map f_c, followed by the retraction $r_c : U_c \to Q_c$, maps the boundary cycle on each omitted face homeomorphically onto that face's boundary. In particular, the classes $(f_c)_*[z_{I,D}]$ span $H_{4c-1}(U_c; \mathbb{F}_2)$.*

Proof. Suppress the fixed coordinates and first use the labels A_i on $q = 4c+1$ points. Let M be the adjacency matrix of the pentagon or rook graph. Their degree and common-neighbor counts give

$$M^2 = c\mathrm{Id} - M + cJ,$$

where J is the all-one matrix. If $Mv = 0$, regularity gives $\mathbf{1}^\top v = 0$, and the displayed identity then gives $cv = 0$. Thus M is invertible. Its rows $\mathbf{1}_{A_i}$ are affinely independent, and their average is $(2c/q)\mathbf{1}$, the barycenter of $\Delta(q, 2c)$. Their convex hull is a full-dimensional simplex containing that barycenter in its interior.

The affine map carries the abstract boundary cycle onto the boundary of this simplex. Every proper face of the omitted coordinate face is retained, so Lemma 9.2(b) identifies r_c here with radial projection. Such a projection maps the boundary of the inner simplex homeomorphically onto the boundary of the coordinate face. For the complementary labels $D \smallsetminus A_i$, replace each row vector by $\mathbf{1} - \mathbf{1}_{A_i}$; the same argument applies. The spanning assertion follows from Lemma 9.3. □

9.6 The short rounding argument for radius one

The following lemma is all the rounding needed for six colors. Its last case is simpler than the corresponding case of the general five-constraint theorem. A reader following radius one may skip Section 9.7 after this proof and continue at Section 9.8.

Lemma 9.6 (Radius-one rounding)**.** *At most five k-sets with a common fractional radius-one center have a common integral center. Thus $\mathcal{C}_1(N,k)$ and Y_1 have the same four-skeleton.*

Proof. Choose a vertex x of the feasible polytope. At most six coordinates are fractional: the sum equality and at most five tight subset inequalities must have full rank on them. Fix the integral coordinates. Let I and O be those fixed to one and zero, and let D be the remaining $m \le 6$ coordinates, of total weight $\ell = k - |I|$. If $m = 0$, we are done. Complementing the coordinates and the original sets if necessary, assume $1 \le \ell \le m/2$.

For a label A, put $T = A \cap D$. Its residual constraint is

$$x(T) \ge k - 1 - |A \cap I| = |T| + |A \cap O| - 1.$$

An empty T gives an automatic constraint. Otherwise $x(T) < |T|$ forces $A \cap O = \varnothing$, and consequently $|T| = k - |A \cap I| \geq \ell$. If T is proper in D, positivity outside T gives $x(T) < \ell$, so $|T| \leq \ell$. If $T = D$, then $\ell \geq m - 1$; together with $\ell \leq m/2$, this forces $(m, \ell) = (2, 1)$, and the constraint follows from $x(D) = \ell$. After discarding automatic constraints, all remaining ones therefore have the form

$$|T| = \ell, \qquad x(T) \geq \ell - 1.$$

We need an ℓ-set meeting every such T in at least $\ell - 1$ points. For $\ell = 1$, any singleton works.

For $\ell = 2$, there are at most five pairs $T_1, \ldots, T_s$. Suppose that no pair meets all of them, and put $H_j = \{i : j \notin T_i\}$ for $j \in D$. These missing-index sets are nonempty and pairwise intersecting. A singleton $H_j = \{i\}$ would force every H_h to contain i, so T_i would be empty. If $H_j = \{i, r\}$, every H_h meets $\{i, r\}$; hence T_i, T_r are disjoint and both miss the positive coordinate j. This would give $x(T_i) + x(T_r) < 2$, a contradiction. Thus $|H_j| \geq 3$ for all j, and

$$s \leq \sum_i x(T_i) = \sum_j x_j(s - |H_j|) \leq 2(s - 3) < s.$$

This contradiction supplies the required pair.

The only remaining possibility is $(m, \ell) = (6, 3)$. Any two constraint triples meet in at least two points: otherwise their scores sum to less than four, because their union has score at most three and their intersection has score less than one. Choose any one of the constraint triples. It meets all the others in at least two points and is the desired integral solution. If there are no constraints, choose any triple. □

9.7 Rounding five subset constraints

The next theorem rounds a fractional center while preserving any fixed coordinates. We need this stronger form for the radius-two comparison. Its proof only involves systems on at most six free coordinates.

Theorem 9.7. *Let $m \geq 1$ and ℓ be integers, let $T_1, \ldots, T_s \subseteq [m]$ with $s \leq 5$, and let $b_1, \ldots, b_s$ be integers. If the polytope*

$$\Pi = \Big\{ x \in [0,1]^m : \sum_j x_j = \ell, \quad x(T_i) \geq b_i \quad (1 \leq i \leq s) \Big\}$$

is nonempty, then it contains a point of $\{0,1\}^m$.

The number five is sharp. On $[4]$, with $\ell = 2$, the six constraints $x(T) \geq 1$ for $T \in \binom{[4]}{2}$ are satisfied by $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, whereas every 2-subset of $[4]$ misses its complementary pair.

Proof. The idea is to pass to a vertex of Π, which has at most six fractional coordinates, and then to treat the three possible shapes of the residual problem by counting. The proof is divided into four steps.

Step 1 (reduction to a strictly fractional vertex). Let x be a vertex of Π, let $I = \{j : x_j = 1\}$, $O = \{j : x_j = 0\}$, and $J = [m] \setminus (I \cup O)$. The restriction $x|_J$ lies in the polytope $\Pi_J \subseteq [0,1]^J$ defined by

$$\sum_{j \in J} y_j = \ell - |I|, \qquad y(T_i \cap J) \geq b_i - |T_i \cap I| \quad (1 \leq i \leq s),$$

a system of the same kind, and a 0/1 point of Π_J, extended by the values of x on $I \cup O$, is a 0/1 point of Π. Moreover $x|_J$ is a vertex of Π_J, since $\Pi_J \times \{x|_{I \cup O}\}$ is a face of Π. At a vertex the tight

constraints have full rank; at $x|_J$ no box constraint is tight, so the equation together with the tight subset inequalities has rank $|J|$. There are at most $s+1 \le 6$ of these, so $|J| \le 6$. (If J is empty, x itself is a 0/1 point.)

It therefore suffices to treat the following situation: $m \le 6$, and Π has a vertex x with $0 < x_j < 1$ for every j. Then, as just observed,

$$\text{the equation and the subset inequalities tight at } x \text{ have rank } m. \tag{9.2}$$

Replacing x by $\mathbf{1} - x$ preserves this situation: the equation becomes $\sum_j y_j = m - \ell$, and $x(T) \ge b$ becomes $y([m] \setminus T) \ge (m-\ell) - |T| + b$, again an integral lower bound; 0/1 points correspond to 0/1 points. Since $0 < \ell < m$, we may therefore assume $1 \le \ell \le m/2$. The remaining cases are $\ell = 1$; $\ell = 2$ with $4 \le m \le 6$; and $(m, \ell) = (6, 3)$.

Step 2 (the case $\ell = 1$). Let $x(T) \ge b$ be one of the constraints. If $T = [m]$ then $b \le x([m]) = 1 = |\{j\} \cap T|$ for every j. If $T \ne [m]$ then $x(T) = 1 - x([m] \setminus T) < 1$, so $b \le 0$. Hence every singleton satisfies all constraints.

Step 3 (the case $\ell = 2$). A constraint with $b \le 0$, or with $T = [m]$ (where $b \le 2$), is satisfied by every 2-subset. For $T \ne [m]$ one has $x(T) < 2$, hence $b \le 1$. Let $T_1, \dots, T_t$ ($t \le 5$) be the sets carrying constraints $x(T_i) \ge 1$ with $T_i \ne [m]$. It remains to find a pair meeting every T_i. Suppose that there is none, and put

$$H_j = \{i : \ j \notin T_i\} \qquad (j \in [m]).$$

A pair $\{j, j'\}$ fails exactly when $H_j \cap H_{j'} \ne \varnothing$. Hence the sets H_j are nonempty (a point j with $H_j = \varnothing$ could be paired with any other point) and pairwise intersecting. We claim that $|H_j| \ge 3$ for every j. If $H_j = \{i\}$, then i belongs to every $H_{j'}$, so $T_i = \varnothing$ and $x(T_i) = 0 < 1$, which is impossible. If $H_j = \{i, i'\}$, then every $H_{j'}$ meets $\{i, i'\}$, so T_i and $T_{i'}$ are disjoint, and neither contains j; hence $x(T_i) + x(T_{i'}) \le 2 - x_j < 2$, which is again impossible. Now count with weights:

$$6 = 3\sum_j x_j \le \sum_j x_j |H_j| = \sum_{i=1}^{t} \bigl(2 - x(T_i)\bigr) \le t \le 5.$$

This is a contradiction.

Step 4 (the case $(m, \ell) = (6, 3)$). Call a constraint $x(T) \ge b$ *trivial* if every 3-subset S of [6] satisfies $|S \cap T| \ge b$, that is, if $b \le \max(0, |T| - 3)$. For $\varnothing \ne T \ne [6]$ the strict inequalities $0 < x_j < 1$ give $x(T) < \min(|T|, 3)$, so $b \le \min(|T|, 3) - 1$. Comparing, one finds that a nontrivial constraint has one of the forms

$$x(e) \ge 1 \ (|e| = 2), \qquad x(T) \ge 1 \ (|T| = 3), \qquad x(T) \ge 2 \ (|T| = 3), \qquad x(T) \ge 2 \ (|T| = 4).$$

First case: some constraint is $x(A) \ge 2$ with $|A| = 3$. We claim that the 3-set A satisfies every constraint. Suppose that $|A \cap T| < b \le x(T)$ for some constraint, and put $a = |A \cap T|$. Since $x(T \setminus A) \le x([6] \setminus A) = 3 - x(A) \le 1$, one has $x(T) < a + 1 \le b$ if $a \ge 1$, using $x(T \cap A) < a$; this is impossible. So $a = 0$, and then $1 \le b \le x(T) \le x([6] \setminus A) \le 1$. Positivity of the coordinates forces $T = [6] \setminus A$, and both constraints $x(A) \ge 2$ and $x([6] \setminus A) \ge 1$ are tight. Their normal vectors add up to $\mathbf{1}$, the normal vector of the equation. Thus the equation and the at most five tight subset inequalities have rank at most five, contradicting (9.2). This proves the claim.

Second case: no such constraint. Count the 3-subsets violating each nontrivial constraint. For $x(e) \ge 1$ they are the four 3-sets disjoint from e; for $x(T) \ge 1$ with $|T| = 3$ it is the single set $[6] \setminus T$; for $x(T) \ge 2$ with $|T| = 4$ they are the four 3-sets containing the pair $e = [6] \setminus T$. There are twenty 3-subsets, so five constraints can exclude all of them only if all five are of the first or the last form

and their excluded sets are pairwise disjoint. Call e a *positive* pair in the first case and a *negative* pair in the last. Two positive pairs with a common point are both avoided by the complement of their union; two negative pairs with a common point are both contained in their union; a positive pair e disjoint from a negative pair e' is avoided by the union of e' and a point outside $e \cup e'$. Hence the positive pairs are pairwise disjoint, the negative pairs are pairwise disjoint, and every positive pair meets every negative pair. Three positive pairs would partition [6], and no pair meets three disjoint pairs; so if there are three positive pairs there is no negative one, and likewise with the signs exchanged. Hence there are at most four such constraints, and some 3-subset satisfies all of them. □

Corollary 9.8. *Let $c \geq 1$, and let σ contain at most five k-subsets of $[N]$. If x is a fractional radius-c center, then σ has an integral center Z with $\mathbf{1}_Z \in F(x)$. In particular, $\mathcal{C}_c(N,k)$ and Y_c have the same four-skeleton.*

Proof. If x is integral, use its support as Z. Otherwise let I be the fixed-one coordinates of $F(x)$ and D its free coordinates. Apply Theorem 9.7 on D, with sum $k - |I|$ and constraints

$$y(A \cap D) \geq k - c - |A \cap I| \qquad (A \in \sigma).$$

They are satisfied by $x|_D$. An integral solution is the indicator of a set $W \subseteq D$; then $Z = I \cup W$ is the required center. □

9.8 Proof of the common topological proposition

We can now prove Proposition 9.1 for both radii, using the radius-two injectivity theorem proved in Section 11. Put $C = \mathcal{C}_c(N,k)$ and $d = 4c - 1$.

For $c = 1$, Lemma 9.6 identifies the four-skeleta of C and Y_1. Their chain groups and boundary maps agree through degree four, so the inclusion is an isomorphism on homology through degree three. For $c = 2$, Theorem 11.1, together with Propositions B.5 and C.8, makes the inclusion injective on homology through degree seven. Both assertions hold over $\mathbb{Z}$ and $\mathbb{F}_2$.

In either case, C and Y_c have the same two-skeleton, so their fundamental groups agree. Since $Y_c \simeq Q_c$ is $(d-1)$-connected, C is simply connected. Its reduced integral homology injects into that of Y_c and hence vanishes through degree $d-1$. Successive applications of the Hurewicz theorem give that C is $(d-1)$-connected.

Finally, Lemma 9.4 makes $(f_c)_*$ injective on $H_d(C;\mathbb{F}_2)$. By Lemma 9.5, the images of the specified cycles span $H_d(U_c;\mathbb{F}_2)$. Given $\xi \in H_d(C;\mathbb{F}_2)$, choose a sum of these cycles with the same image as ξ. Their difference has zero image and is therefore zero. This proves the generation assertion.

10 The uniform box-complex index bound

We prove Theorem 1.3. The upper bound comes from an explicit sphere map and does not use the local homology calculations. Matching lower bounds for $r = 2, 3$ follow from Proposition 9.1. Write $c = r - 1$, and assume $n \geq 2c + 1$.

10.1 The index bound and its meaning

Use the weighted-pair model $B(G) = \mathrm{Hom}(K_2, G)$ from Section 9.1. A map to a sphere is *equivariant* if exchanging the two sides sends its value to its negative. This model and the simplicial box complex in Matoušek's book are $\mathbb{Z}_2$-homotopy equivalent [Cso08, Theorem 9, p. 539], so their indices agree.

Lemma 10.1 (The sphere map)**.** *For every $c \geq 1$ and $n \geq 2c+1$, there is an explicit equivariant map*

$$B(G_{n,c+1}) \longrightarrow S^{4c-1}.$$

Consequently, $\operatorname{ind}_{\mathbb{Z}_2} B(G_{n,c+1}) \leq 4c-1$*, and* $\mathcal{N}(G_{n,c+1})$ *is not* $(4c-1)$*-connected.*

The restriction $n \geq 2c+1$ matters: $G_{2,2} = K_6$ has box complex equivariantly homotopy equivalent to S^4, hence index four [BK06, Proposition 4.3].

Within the hierarchy of topological chromatic bounds [MZ04], the usual index bound is

$$\chi(G) \geq \operatorname{ind}_{\mathbb{Z}_2} B(G) + 2 \qquad \text{[Mat03, Theorem 5.9.3, p. 130].}$$

Lemma 10.1 shows that this bound can certify at most $4c+1$ colors in our range of n. An upper bound on the index does not itself supply a chromatic lower bound. For $c = 1, 2$, however, the connectivity proved in Proposition 9.1 gives the matching index lower bound and hence the exact values

$$\operatorname{ind}_{\mathbb{Z}_2} B(G_{n,c+1}) = 4c-1, \qquad \operatorname{conn} \mathcal{N}(G_{n,c+1}) = 4c-2 \qquad (c = 1, 2). \tag{10.1}$$

Here $\operatorname{conn} X$ is the largest integer t for which X is t-connected.

10.2 Reducing to a fixed number of coordinates

First we reduce n to $2c+1$. A graph homomorphism is a map on vertices that preserves adjacency. Such a map induces an equivariant map on $B(G)$: transfer the weight at each vertex to its image, adding weights when several vertices have the same image. The two resulting supports still form a complete bipartite subgraph. We use the deletion map behind Jafari and Alipour's monotonicity inequality [JA17, Theorem 3.1, p. 73], including its proof here.

Lemma 10.2 (Balanced deletion)**.** *For $n \geq m \geq c+1$, there is a graph homomorphism $G_{n,c+1} \to G_{m,c+1}$.*

Proof. It suffices to reduce k to $k-1$ when $k \geq c+2$. Delete two coordinates a, b from $[2k]$, and order the remaining set U. For a k-set A, define a $(k-1)$-subset of U by

$$p(A) = \begin{cases} A \cap U, & |A \cap \{a,b\}| = 1, \\ (A \cap U) \cup \{\min(U \setminus A)\}, & |A \cap \{a,b\}| = 2, \\ A \setminus \{\min A\}, & |A \cap \{a,b\}| = 0. \end{cases}$$

We claim that $|p(A) \cap p(C)| \leq |A \cap C|$ for all A, C. If neither set contains both deleted coordinates, the map only removes elements. If both contain them, their intersection loses two elements and gains at most two. If one contains both and the other contains exactly one, it loses one and gains at most one.

The only remaining case is that A contains both and C contains neither. Put $u = \min(U \setminus A)$ and $v = \min C$. A new common element is possible only if $u \in C$, in which case $v \leq u$. If $v = u$, that element is deleted from C. If $v < u$, the definition of u gives $v \in A$, so deleting v removes an old common element and offsets the possible gain. This proves the claim. Adjacent inputs therefore have images intersecting in at most c elements. The images cannot coincide, since each has $k-1 > c$ elements. Iterating proves the lemma. □

It is now enough to construct the sphere map for $G_{2c+1,c+1}$ and compose it with the map induced by Lemma 10.2. This reduces the construction to $4c+2$ coordinates, independently of the original ground-set size.

10.3 A sphere map from averaged indicator vectors

Set $m = 2c+1$ and $N = 2m = 4c+2$. At a point (λ, μ) of $B(G_{m,c+1})$, average the indicator vectors on each side:

$$x = \sum_A \lambda_A \mathbf{1}_A, \qquad y = \sum_C \mu_C \mathbf{1}_C.$$

The sums range over the m-subsets of $[N]$. Every coordinate lies in $[0,1]$, and $\sum_j x_j = \sum_j y_j = m$. Moreover, every pair carrying positive weights is an edge, so

$$x \cdot y = \sum_{A,C} \lambda_A \mu_C |A \cap C| \le c. \tag{10.2}$$

Thus the averaged vectors cannot be too close. To quantify this, put $v = x - y$ and $u = (x+y)/2$. Since $\sum_j u_j = m$, Cauchy–Schwarz gives $\|u\|^2 \ge m^2/N = m/2 = c + 1/2$. Together with (10.2), this yields

$$\|v\|^2 = 4(\|u\|^2 - x \cdot y) \ge 2. \tag{10.3}$$

All norms here are Euclidean.

The coordinates of v sum to zero. Normalizing v would therefore give a map to S^{N-2}, one dimension above our target. The quantitative separation in (10.3) lets us remove one more dimension: discard the first coordinate and subtract the mean of the remaining coordinates. Explicitly, put

$$w_j = v_j + \frac{v_1}{N-1} \quad (2 \le j \le N).$$

Then $\sum_{j=2}^N w_j = 0$, and expanding the squares gives

$$\|w\|^2 = \|v\|^2 - \frac{N}{N-1} v_1^2 \ge 2 - \frac{N}{N-1} = \frac{N-2}{N-1} > 0,$$

where we used $|v_1| \le 1$. We may therefore define

$$\Phi(\lambda, \mu) = \frac{w}{\|w\|}.$$

This is a continuous map to the unit sphere of $\{w \in \mathbb{R}^{N-1} : \sum_j w_j = 0\}$, a vector space of dimension $N-2$. Its unit sphere is $S^{N-3} = S^{4c-1}$. Exchanging λ and μ exchanges x and y, hence sends v, w, and Φ to their negatives. The map is equivariant, proving the index assertion of Lemma 10.1.

For the connectivity assertion, recall that $B(G)$ and $\mathcal{N}(G)$ are homotopy equivalent as ordinary spaces. A t-connected free $\mathbb{Z}_2$-space has index at least $t+1$ [Mat03, Proposition 5.3.2(iv), pp. 96–98]; Section 9.1 described the successive extensions over hemispheres that prove this fact. If $\mathcal{N}(G_{n,c+1})$ were $(4c-1)$-connected, its box complex would have index at least $4c$, contradicting the map just constructed. This completes the proof of Lemma 10.1 and, using Proposition 9.1, of (10.1). This proves Theorem 1.3.

Where the extension to larger c remains open. The polyhedral deformation of Lemma 9.2 holds for every c: its retained-face model Q_c is $(4c-2)$-connected. Transferring this conclusion to the neighborhood complex requires the separate discrete-to-polyhedral comparison, which we have proved only for $c = 1, 2$. Even a matching connectivity theorem for larger c would give only $4c+1$ colors by the standard bound. To obtain $4c+2$ by our method would additionally require suitable homology generators and local coloring constraints that kill their images.

11 Integral and fractional radius-two centers

Fix $1 \le k < N$. Throughout this section the radius is two, and we abbreviate $\mathcal{C} = \mathcal{C}_2(N,k)$, $Y = Y_2$, $U = U_2$, $Q = Q_2$, $P_Z = P_{2,Z}$, $P_\sigma = \bigcap_{A\in\sigma} P_A$, $i = i_2$, and $f = f_2$. Coefficients $\Bbbk$ are either $\mathbb{Z}$ or $\mathbb{F}_2$. Reduced homology in degree -1 vanishes exactly when a complex is nonempty; groups in smaller degrees are zero.

The goal is injectivity of the inclusion on homology through degree seven. Its two finite inputs are proved in Appendices B and C.

Theorem 11.1. *Suppose that*

$$\widetilde{H}_t(\mathcal{C}_2(9,3);\mathbb{Z}) = \widetilde{H}_t(\mathcal{C}_2(8,4);\mathbb{Z}) = 0 \qquad (0 \le t \le 7). \tag{11.1}$$

Then $i_*\colon H_j(\mathcal{C};\Bbbk) \to H_j(Y;\Bbbk)$ is injective for $0 \le j \le 7$.

Remark. The idea of the proof is to record, together with each simplex of labels, a simplex of centers. Three things are needed. A cycle of $\mathcal{C}$ must be liftable to such recorded data, with centers chosen compatibly on common faces; this requires the spaces of admissible centers of a family to be acyclic in a range of degrees that shrinks as the family grows (Lemma 11.9). A boundary in Y must lift to a boundary of recorded data; here all fractional centers are admissible and form a contractible set (Lemma 11.8). Finally, recorded data must be converted back into chains of $\mathcal{C}$ without altering a lifted cycle (Lemma 11.10); this requires certain local complexes to be acyclic through degree seven, and it is here that (11.1) enters. Because injectivity in degree seven concerns fillings of dimension eight, the conversion must be defined through degree eight.

11.1 Hulls of integral centers

For families of six or more labels a fractional center need not round to an integral one. In the comparison argument such families will instead be equipped with a space of centers built from their integral centers only. The mediator below will fill every triangle of integral centers.

For a face σ of $\mathcal{C}_2(N,k)$ let $\mathcal{Z}_\sigma$ be the set of its centers, and define

$$T_\sigma = \bigcup_{F \text{ retained}} \operatorname{conv}\{\mathbf{1}_Z : \ Z \in \mathcal{Z}_\sigma,\ \mathbf{1}_Z \in F\},$$

the union over the retained faces F of the convex hulls of those integral centers which lie in F (empty hulls are omitted). Thus $T_\sigma \subseteq Q$ is a finite union of polytopes. For a nonempty face σ of Y we also put $D_\sigma = P_\sigma \cap Q$, the set of fractional centers lying in Q.

Lemma 11.2 (Segments of centers)**.** *Let σ be a face of $\mathcal{C}_2(N,k)$.*

(a) Every $x \in T_\sigma$ lies in the convex hull of the integral centers of σ lying in $F(x)$. In particular $F(x)$ contains an integral center of σ.

(b) $T_\sigma \subseteq D_\sigma$, and $T_\sigma \subseteq T_\tau$ whenever $\varnothing \ne \tau \subseteq \sigma$.

(c) If $Z, Z' \in \mathcal{Z}_\sigma$, then $d_J(Z,Z') \le 4$, and the segment from $\mathbf{1}_Z$ to $\mathbf{1}_{Z'}$ lies in T_σ.

Proof. (a) Write $x = \sum \lambda_Z \mathbf{1}_Z$ with $\lambda_Z > 0$, the Z being centers in a retained face. A coordinate in which x equals 0 or 1 has the same value in every $\mathbf{1}_Z$ occurring, so all of them lie in $F(x)$. (b) Each $\mathbf{1}_Z$ with $Z \in \mathcal{Z}_\sigma$ lies in the convex set P_σ, and $\mathcal{Z}_\sigma \subseteq \mathcal{Z}_\tau$. (c) For $A \in \sigma$ one has $d_J(Z,Z') \le d_J(Z,A) + d_J(A,Z') \le 4$. The smallest face containing $\mathbf{1}_Z$ and $\mathbf{1}_{Z'}$ has the $2d_J(Z,Z') \le 8$ coordinates of $Z \triangle Z'$ as free coordinates, so it is retained and contains the segment. □

The essential point is that every triangle of integral centers can be filled inside T_σ. This rests on the following lemma.

Lemma 11.3 (Mediator)**.** *Let Z_1, Z_2, Z_3 be k-sets which have a common center. There is a k-set W such that*

(1) $d_J(W, A) \le 2$ for every k-set A which is a center of $\{Z_1, Z_2, Z_3\}$;

(2) for each pair i, j, the smallest face containing $\mathbf{1}_W$, $\mathbf{1}_{Z_i}$ and $\mathbf{1}_{Z_j}$ is retained.

Proof. If the smallest face containing $\mathbf{1}_{Z_1}, \mathbf{1}_{Z_2}, \mathbf{1}_{Z_3}$ is retained, then $W = Z_1$ has both properties. Suppose it is omitted. We construct W in three steps.

Step 1 (blocks). Divide the coordinates according to their membership in the three sets:

block	points belonging to	size
C	all three sets	$\lvert C\rvert$
X_i	Z_i only	x_i
Y_i	the two sets other than Z_i	y_i
E	none of the sets	$\lvert E\rvert$

The free coordinates of the face spanned by the three sets are those in the six blocks X_i, Y_i. Since Z_i consists of C, X_i and the two blocks Y_j with $j \ne i$, and all three sets have k elements, there is an integer t with

$$x_i = y_i + t \ (i = 1, 2, 3), \qquad \ell := k - |C| = y_1 + y_2 + y_3 + t, \qquad m := \sum_i (x_i + y_i) = 2\ell + t.$$

The face spanned by the three sets is of type $\Delta(m, \ell)$. Complementing all sets interchanges C with E and X_i with Y_i, replaces t by $-t$, and preserves distances and retention; so we may assume $t \ge 0$. Relabel so that $y_3 = \max(y_1, y_2, y_3)$. Since $0 \le t \le \ell$, one has $\rho = \ell(\ell + t)/(2\ell + t) \le 2\ell/3$, and the face is omitted only if $\ell \ge 4$.

Step 2 (a median). Give each coordinate the weight w_q equal to the number of the sets Z_i containing it. For a k-set A put $S(A) = \sum_i d_J(A, Z_i) = 3k - \sum_{q \in A} w_q$. Let W consist of C, the three blocks Y_i, and t points of X_3 (possible since $x_3 = y_3 + t$). Then $|W| = |C| + (\ell - t) + t = k$, the set W consists of k coordinates of largest weight, and

$$S(W) = 3\ell - 2(\ell - t) - t = \ell + t$$

is the least value of S. A center A of the three sets has $S(A) \le 6$; hence $\ell + t \le 6$, and $t \le 2$ because $\ell \ge 4$. Now let A be any center of the three sets, and $p = d_J(A, W)$. Among the p points of $W \smallsetminus A$ at most t have weight one and the others have weight at least two, while the p points of $A \smallsetminus W$ have weight at most one. Hence

$$S(A) - S(W) \ge (2p - t) - p = p - t,$$

and $p \le S(A) - S(W) + t \le 6 - \ell \le 2$. This proves (1).

Step 3 (the three triangles). Let $\{h, i, j\} = \{1, 2, 3\}$. The pair Z_i, Z_j has distance $L_h = x_i + y_j = \ell - y_h$, which is at most four because both sets lie at distance at most two from a common center. It spans a face of type $\Delta(2L_h, L_h)$. Adding $\mathbf{1}_W$ creates a new free coordinate only where Z_i and Z_j agree and W differs from them. This happens only for $h = 3$, at the t chosen points of X_3. Hence the three faces in (2) have $2L_1$, $2L_2$ and $2L_3 + t$ free coordinates. If $t = 0$ all of these are at most eight. If $t > 0$ then L_3 is the smallest of the three distances, and $3L_3 \le L_1 + L_2 + L_3 = 2\ell + t \le 12 - t < 12$; so $L_3 \le 3$ and $2L_3 + t \le 8$. A face with at most eight free coordinates is retained. □

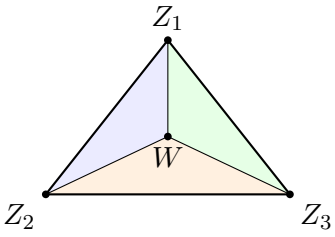


Figure 7: The mediator supplies a simplicial filling of a triangle of centers. This is an incidence diagram: the three filled triangles lie in retained coordinate faces, even when the original three-center face is omitted.

Proposition 11.4 (Integral center hulls)**.** *For every face σ of $\mathcal{C}_2(N,k)$ the space T_σ is nonempty and simply connected. In particular $\widetilde{H}_0(T_\sigma;\Bbbk) = H_1(T_\sigma;\Bbbk) = 0$ for $\Bbbk = \mathbb{Z}, \mathbb{F}_2$.*

Proof. A center Z of σ gives the point $\mathbf{1}_Z \in T_\sigma$. By Lemma 11.2(a) every point x of T_σ is joined by a segment inside $\operatorname{conv}(\mathcal{Z}_\sigma \cap F(x)) \subseteq T_\sigma$ to a center Z_x in $F(x)$, and by Lemma 11.2(c) any two centers are joined by a segment in T_σ. Hence T_σ is path connected.

Triangulate T_σ so that each of the polytopes $\operatorname{conv}(\mathcal{Z}_\sigma \cap F)$ is a subcomplex. By simplicial approximation every loop is homotopic to an edge loop [Hat02, Theorem 2C.1]. For each vertex x of the triangulation choose a center Z_x in $F(x)$. An edge $[x,y]$ lies in some $\operatorname{conv}(\mathcal{Z}_\sigma \cap F)$; then $F(x)$ and $F(y)$ are faces of F, so $\mathbf{1}_{Z_x}$ and $\mathbf{1}_{Z_y}$ lie in the same convex set, and straight-line interpolation moves the edge onto the segment $[\mathbf{1}_{Z_x}, \mathbf{1}_{Z_y}]$ inside T_σ. Thus every loop is homotopic to a closed polygon whose vertices are centers $Z^{(0)}, Z^{(1)}, \dots, Z^{(p)} = Z^{(0)}$ and whose sides are the segments of Lemma 11.2(c).

Using the segments from $Z^{(0)}$ to the other vertices, such a polygon is a product of triangular loops $Z^{(0)} \to Z \to Z' \to Z^{(0)}$, so it suffices to show that each of these is null-homotopic. Every label of σ is a common center of $Z^{(0)}, Z, Z'$. Let W be the mediator of Lemma 11.3. By property (1), every label of σ lies at distance at most two from W, so $W \in \mathcal{Z}_\sigma$. By property (2), the three triangles $\operatorname{conv}(\mathbf{1}_W, \mathbf{1}_{Z^{(0)}}, \mathbf{1}_Z)$, $\operatorname{conv}(\mathbf{1}_W, \mathbf{1}_Z, \mathbf{1}_{Z'})$ and $\operatorname{conv}(\mathbf{1}_W, \mathbf{1}_{Z'}, \mathbf{1}_{Z^{(0)}})$ lie in retained faces, hence in T_σ. Mapping the cone on the boundary of a triangle affinely onto these three triangles gives a null-homotopy of the loop. Hence T_σ is simply connected. The statement about homology follows from the Hurewicz isomorphism in degree one and the universal coefficient theorem. □

11.2 Local complexes of centers

For a retained face F let $\mathcal{K}_F$ be the subcomplex of $\mathcal{C}$ consisting of the families having a center among the integral vertices of F; the labels themselves need not lie in F. Clearly $\mathcal{K}_F \subseteq \mathcal{K}_G$ when $F \subseteq G$. Let $\mathcal{C}_2(F) \subseteq \mathcal{K}_F$ be the subcomplex of families of integral vertices of F having a center among the integral vertices of F. Deleting the fixed coordinates preserves Johnson distances, so $\mathcal{C}_2(F)$ is isomorphic to $\mathcal{C}_2(m,\ell)$ when F is of type $\Delta(m,\ell)$.

Lemma 11.5 (Coordinate-face retractions)**.** *For every face F of $\Delta(N,k)$ there is a map p_F from $\binom{[N]}{k}$ onto the set of integral vertices of F which fixes the integral vertices of F and satisfies $d_J(p_F(A), p_F(B)) \le d_J(A,B)$ for all A, B.*

Proof. We fix the coordinates of $I \cup O$ one at a time, and it suffices to treat one step. Let F' be a face with free set D', let $j \in D'$, and suppose the face $F'' = F' \cap \{x_j = 0\}$ is nonempty. Order D', and for an integral vertex Z of F' define $p_j(Z) = Z$ if $j \notin Z$, and $p_j(Z) = (Z \smallsetminus \{j\}) \cup \{a(Z)\}$ if

$j \in Z$, where $a(Z)$ is the least element of $D' \smallsetminus (Z \cup \{j\})$; it exists because F'' is nonempty. We claim $|p_j(Z) \cap p_j(Z')| \geq |Z \cap Z'|$. This is clear if neither set contains j, and if exactly one does, since then j is not a common element. If both contain j, the common element j is lost; if $a(Z) = a(Z')$ it is replaced by this element, and if, say, $a(Z) < a(Z')$ then $a(Z) \in Z'$ by minimality of $a(Z')$, so $a(Z)$ becomes a common element. Thus p_j is 1-Lipschitz, maps onto the integral vertices of F'', and fixes them. Fixing j to the value 1 is the same operation applied to complements, complementation being an isometry from k-sets to $(N-k)$-sets. Composing these maps, starting from $F' = \Delta(N,k)$, gives p_F. □

Lemma 11.6 (Local acyclicity)**.** *Assume* (11.1). *Then* $\widetilde{H}_t(\mathcal{K}_F; \Bbbk) = 0$ *for every retained face* F *and* $-1 \leq t \leq 7$.

Proof. Step 1. The inclusion $\mathcal{C}_2(F) \subseteq \mathcal{K}_F$ is a homotopy equivalence. Indeed, if Z is an integral vertex of F and a center of $\sigma \in \mathcal{K}_F$, then $p_F(Z) = Z$ is a center of $p_F(\sigma)$ and of $\sigma \cup p_F(\sigma)$, by Lemma 11.5. So p_F is a simplicial retraction of $\mathcal{K}_F$ onto $\mathcal{C}_2(F)$, and its composite with the inclusion is contiguous to the identity, hence homotopic to it [Spa66, §3.5].

Step 2. Let F be of type $\Delta(m,\ell)$. If $\min(\ell, m-\ell) \leq 2$, any two integral vertices of F are at distance at most two, and $\mathcal{C}_2(F)$ is a full simplex. Complementation within F gives $\mathcal{C}_2(m,\ell) \cong \mathcal{C}_2(m, m-\ell)$, so let $3 \leq \ell \leq m/2$. The condition $\ell(m-\ell) \leq 2m$ then leaves $\ell = 3$ with $6 \leq m \leq 9$, and $(m,\ell) = (8,4)$; for $\ell \geq 5$ one has $\ell(m-\ell) - 2m = (\ell-2)m - \ell^2 \geq \ell^2 - 4\ell > 0$.

Step 3. For $m \leq 8$ let F_m be the face of $\Delta(9,3)$ with fixed-zero set $\{m+1, \dots, 9\}$. By Lemma 11.5, p_{F_m} maps every face of $\mathcal{C}_2(9,3)$ to a face of $\mathcal{C}_2(F_m) \cong \mathcal{C}_2(m,3)$, and it fixes $\mathcal{C}_2(F_m) \subseteq \mathcal{C}_2(9,3)$. So $\mathcal{C}_2(m,3)$ is a retract of $\mathcal{C}_2(9,3)$, and its reduced homology is a direct summand of that of $\mathcal{C}_2(9,3)$. Together with (11.1) this gives $\widetilde{H}_t(\mathcal{C}_2(F); \mathbb{Z}) = 0$ for $t \leq 7$ in all cases, and the universal coefficient theorem gives the same over $\mathbb{F}_2$.

□

11.3 Choosing admissible center spaces

For a nonempty face σ of Y recall that $D_\sigma = P_\sigma \cap Q$. The deformation of Lemma 9.2(b) maps the nonempty convex set P_σ into itself and ends in D_σ, which it fixes; hence D_σ is contractible. For a face σ of $\mathcal{C}$ define

$$E_\sigma = \begin{cases} D_\sigma, & |\sigma| \leq 5, \\ T_\sigma, & |\sigma| \geq 6. \end{cases}$$

Lemma 11.7 (Admissible centers)**.** *Let* σ *be a face of* $\mathcal{C}$.

(a) $E_\sigma \subseteq D_\sigma$; *if* $\varnothing \neq \tau \subseteq \sigma$ *then* $D_\sigma \subseteq D_\tau$ *and* $E_\sigma \subseteq E_\tau$.

(b) $\widetilde{H}_t(E_\sigma; \Bbbk) = 0$ *for* $-1 \leq t \leq 7 - |\sigma|$.

(c) For every $x \in E_\sigma$ *the face* $F(x)$ *contains an integral center of* σ.

Proof. (a) Use Lemma 11.2(b); when $|\tau| \leq 5 < |\sigma|$ one has $E_\sigma = T_\sigma \subseteq D_\sigma \subseteq D_\tau = E_\tau$. (b) For $|\sigma| \leq 5$ the set D_σ is contractible. For $|\sigma| \geq 6$ the range is $t \leq 1$, and Proposition 11.4 applies. (c) For $|\sigma| \geq 6$ this is Lemma 11.2(a); for $|\sigma| \leq 5$ it is Corollary 9.8. □

The required range shrinks with the number of labels:

$\lvert\sigma\rvert$	Chosen space	Property used in a seven-cycle lift
at most 5	D_σ	contractible; five-label rounding applies
6	T_σ	connected, with $H_1 = 0$
7	T_σ	connected
8	T_σ	nonempty

This is why simple connectivity of the integral center hulls suffices.

Part (c) is the bridge between the two kinds of centers: a point of E_σ certifies that σ belongs to $\mathcal{K}_{F(x)}$.

Choose a triangulation $\mathfrak{T}$ of Q in which every retained face, every D_σ ($\sigma \in Y$) and every T_σ ($\sigma \in \mathcal{C}$) is a subcomplex; it exists because these are finitely many finite unions of polytopes [RS72, Chapter 2]. Every simplex a of $\mathfrak{T}$ lies in a retained face; let $F(a)$ be the smallest face of $\Delta(N,k)$ containing a. It is retained, and $F(a') \subseteq F(a)$ for every face a' of a; for a vertex x of $\mathfrak{T}$ this agrees with the earlier notation $F(x)$.

11.4 Keeping centers with chains

Let $V = \binom{[N]}{k}$. Let $C_*(\Delta^V)$ be the oriented simplicial chain complex of the full simplex on V, and $C_*(\mathfrak{T})$ that of $\mathfrak{T}$, both with coefficients in $\Bbbk$ and without augmentation. In the tensor product, with differential

$$\partial(\sigma \otimes a) = \partial\sigma \otimes a + (-1)^{\dim\sigma}\sigma \otimes \partial a,$$

let B be the span of the elements $\sigma \otimes a$ with $\sigma \in Y$ and a a simplex of D_σ, and let $B^E \subseteq B$ be the span of those with $\sigma \in \mathcal{C}$ and $a \subseteq E_\sigma$. Since $D_\sigma \subseteq D_\tau$ for nonempty $\tau \subseteq \sigma \in Y$, and by Lemma 11.7(a), both are subcomplexes. A generator $\sigma \otimes a$ records a family of labels together with a simplex of admissible centers for it; its degree is $\dim\sigma + \dim a$, and we call $\dim\sigma$ its *label dimension*. (Geometrically, B is the cellular chain complex of $\bigcup_{\sigma\in Y} |\sigma| \times D_\sigma$.)

Let $\mathrm{aug}\colon C_*(\mathfrak{T}) \to \Bbbk$ be the augmentation, which sends each vertex to 1. The chain maps $\mathrm{id} \otimes \mathrm{aug}$ give

$$\varepsilon_Y\colon B \to C_*(Y), \qquad \varepsilon_{\mathcal{C}}\colon B^E \to C_*(\mathcal{C}), \qquad \sigma \otimes a \mapsto \begin{cases} \sigma, & \dim a = 0, \\ 0, & \dim a > 0, \end{cases}$$

and $\varepsilon_Y = i \circ \varepsilon_{\mathcal{C}}$ on B^E.

Lemma 11.8 (Filling recorded data)**.** *If $b \in B$ is a cycle and $\varepsilon_Y(b)$ is a boundary in $C_*(Y)$, then b is a boundary in B.*

Proof. Suppose first that $\varepsilon_Y(b) = 0$, and let p be the largest label dimension occurring in b. Write the part of b of label dimension p as $\sum_\sigma \sigma \otimes c_\sigma$, with $\dim\sigma = p$ and c_σ a chain of D_σ. The terms of ∂b of label dimension p are $(-1)^p \sum_\sigma \sigma \otimes \partial c_\sigma$, so each c_σ is a cycle. If c_σ has degree zero, the vanishing of $\varepsilon_Y(b)$ says that its augmentation is zero. Since D_σ is contractible, $c_\sigma = \partial d_\sigma$ for a chain d_σ of D_σ. Now

$$b - \partial\Big(\sum_\sigma (-1)^p \sigma \otimes d_\sigma\Big) = b - \sum_\sigma \sigma \otimes c_\sigma - \sum_\sigma (-1)^p\, \partial\sigma \otimes d_\sigma$$

is a cycle with no terms of label dimension p or more, and its image under ε_Y is still zero, since every d_σ has positive degree. Repeating this, one reaches zero, so b is a boundary.

In general, write $\varepsilon_Y(b) = \partial v$. Choosing for each simplex σ occurring in v a vertex of D_σ produces $u \in B$ with $\varepsilon_Y(u) = v$. Then $b - \partial u$ is a cycle with $\varepsilon_Y(b - \partial u) = 0$, and the first case applies. □

Lemma 11.9 (Lifting an integral cycle)**.** *Let $0 \le j \le 7$. For every j-cycle z of $\mathcal{C}$ there is a cycle $\tilde{z} \in B^E$ of degree j with $\varepsilon_{\mathcal{C}}(\tilde{z}) = z$.*

Proof. Write $z = \sum_\sigma c_\sigma \sigma$ with $\dim \sigma = j$. Since $|\sigma| = j+1 \le 8$, Lemma 11.7(b) shows that E_σ is nonempty; choose a vertex x_σ of $\mathfrak{T}$ in E_σ, and put $\tilde{z}_0 = \sum_\sigma c_\sigma \, \sigma \otimes x_\sigma$. Then $\varepsilon_{\mathcal{C}}(\tilde{z}_0) = z$, and $\partial \tilde{z}_0$ has label dimension $j-1$.

Suppose that for some $s \ge 0$ a chain $\tilde{z}_s \in B^E$ of degree j has been found with $\varepsilon_{\mathcal{C}}(\tilde{z}_s) = z$ and with $\partial \tilde{z}_s$ of label dimension at most $p = j - s - 1$. If $p < 0$ we are done. Otherwise write the part of $\partial \tilde{z}_s$ of label dimension p as $\sum_\tau \tau \otimes w_\tau$, with $\dim \tau = p$ and w_τ a chain of degree s in E_τ. The terms of label dimension p in $0 = \partial\partial \tilde{z}_s$ are $(-1)^p \sum_\tau \tau \otimes \partial w_\tau$, so each w_τ is a cycle. For $s = 0$ its augmentation is the coefficient of τ in $\partial \varepsilon_{\mathcal{C}}(\tilde{z}_0) = \partial z = 0$. Since $s = j - 1 - p \le 6 - p = 7 - |\tau|$, Lemma 11.7(b) gives a chain y_τ of E_τ with $\partial y_\tau = w_\tau$. Put

$$\tilde{z}_{s+1} = \tilde{z}_s - (-1)^p \sum_\tau \tau \otimes y_\tau .$$

Since $\partial(\tau \otimes y_\tau) = \partial\tau \otimes y_\tau + (-1)^p \tau \otimes w_\tau$, the terms of label dimension p cancel in $\partial \tilde{z}_{s+1}$, and the new terms have label dimension $p-1$ (they vanish if $p = 0$, the chain complexes being unaugmented). The image under $\varepsilon_{\mathcal{C}}$ is unchanged. After at most $j+1$ steps one obtains the required cycle. □

Lemma 11.10 (Returning to integral chains)**.** *Assume* (11.1). *There are $\Bbbk$-linear maps $\phi\colon B_j \to C_j(\mathcal{C})$ for $0 \le j \le 8$ such that*

(i) $\partial\phi = \phi\partial$ on B_j for $1 \le j \le 8$;

(ii) $\phi = \varepsilon_{\mathcal{C}}$ on B_j^E;

(iii) $\phi(\sigma \otimes a)$ is a chain of $\mathcal{K}_{F(a)}$ for every generator $\sigma \otimes a$.

Proof. We define ϕ on generators by induction on the degree. On B^E we set $\phi = \varepsilon_{\mathcal{C}}$. This is compatible with (iii): a generator $\sigma \otimes a$ of B^E with $\dim a > 0$ is sent to 0, and a generator $\sigma \otimes x$ with x a vertex is sent to σ, which belongs to $\mathcal{K}_{F(x)}$ by Lemma 11.7(c). It is compatible with (i), because $\varepsilon_{\mathcal{C}}$ is a chain map and B^E is a subcomplex.

Every generator of degree zero has the form $\{A\} \otimes x$ with $x \in D_{\{A\}} = E_{\{A\}}$, so it lies in B^E and is sent to the vertex A. Now let $1 \le n \le 8$, suppose ϕ has been defined in degrees below n, and let $b = \sigma \otimes a$ be a generator of degree n not in B^E. The chain $\phi(\partial b)$ lies in $\mathcal{K}_{F(a)}$: the terms $\tau \otimes a$ of ∂b are sent into $\mathcal{K}_{F(a)}$, and the terms $\sigma \otimes a'$, with a' a facet of a, into $\mathcal{K}_{F(a')} \subseteq \mathcal{K}_{F(a)}$. It is a cycle: for $n \ge 2$ because $\partial\phi(\partial b) = \phi(\partial\partial b) = 0$ by (i) in degree $n-1$, and for $n = 1$ because ϕ sends every generator of degree zero to a single vertex, so that $\phi(\partial b)$ has augmentation zero. By Lemma 11.6, $\widetilde{H}_{n-1}(\mathcal{K}_{F(a)}; \Bbbk) = 0$, so we may choose $\phi(b) \in C_n(\mathcal{K}_{F(a)})$ with $\partial\phi(b) = \phi(\partial b)$. This completes the induction and the proof. □

11.5 Proof of Theorem 11.1

Let $0 \le j \le 7$, and let z be a j-cycle of $\mathcal{C}$ whose image is a boundary in Y, say $i(z) = \partial v$. By Lemma 11.9 there is a cycle $\tilde{z} \in B_j^E$ with $\varepsilon_{\mathcal{C}}(\tilde{z}) = z$. Then $\varepsilon_Y(\tilde{z}) = i(z) = \partial v$, and Lemma 11.8 gives $w \in B_{j+1}$ with $\tilde{z} = \partial w$. Since $j+1 \le 8$, Lemma 11.10 gives

$$z = \varepsilon_{\mathcal{C}}(\tilde{z}) = \phi(\tilde{z}) = \phi(\partial w) = \partial\phi(w).$$

Thus z is a boundary in $\mathcal{C}$.

A Filling from fibers

The two local computations rest on one chain-level device, which we now describe, together with two standard facts about nerves. In this section and the next two, homology has integer coefficients, and chain complexes of simplicial complexes are *augmented*: the empty simplex $[\varnothing]$ spans degree -1, and $\partial[v] = [\varnothing]$ for a vertex v. Thus $\widetilde{H}_{-1}(L) = 0$ means that L has at least one vertex.

Let K be a finite simplicial complex whose vertex set is divided into *base* vertices and *fiber* vertices. For a face R of K consisting of base vertices ($R = \varnothing$ allowed), the **fiber** of R is the simplicial complex

$$K_R = \{S : \ S \text{ is a set of fiber vertices and } R \cup S \in K\},$$

which always contains the empty set.

Lemma A.1 (Filling from fibers)**.** *Let $d \geq 0$. Suppose that $\widetilde{H}_t(K_R) = 0$ whenever R is a face of base vertices and $-1 \leq t \leq d - |R|$. Then $\widetilde{H}_j(K) = 0$ for $0 \leq j \leq d$.*

Proof. Order the vertices of K so that the base vertices come first. For an ordered face $[R]$ of base vertices and a chain c of K_R, let $[R] * c$ be the chain obtained by placing the vertices of R in front of those of each simplex of c, with $[R] * [\varnothing] = [R]$. For $|R| \geq 1$ one has

$$\partial([R] * c) = (\partial[R]) * c + (-1)^{|R|}[R] * \partial c,$$

where $(\partial[R]) * c$ is formed in the same way and $[\varnothing] * c = c$.

Let z be a j-cycle of K with $0 \leq j \leq d$, and let p be the largest number of base vertices of a simplex occurring in z. Collect the simplices with exactly p base vertices according to their base part: this part of z is $\sum_{|R|=p}[R] * z_R$, where z_R is a chain of K_R of degree $j - p \geq -1$. In ∂z the simplices with p base vertices are exactly those of $(-1)^p \sum_R [R] * \partial z_R$; hence each z_R is a cycle. Since $-1 \leq j - p \leq d - p$, the hypothesis gives chains b_R of K_R with $\partial b_R = z_R$, and

$$z - \partial \sum_{|R|=p} (-1)^p [R] * b_R = z - \sum_R [R] * z_R - \sum_R (-1)^p (\partial[R]) * b_R$$

is a cycle of K (every simplex of $[R] * b_R$ belongs to K by the definition of K_R) all of whose simplices have fewer than p base vertices. Repeating, one reaches a cycle of $K_\varnothing$ of degree $j \leq d$, which bounds by hypothesis. Hence z is a boundary. □

The fibers that occur will be unions of full simplices. For a finite set S let $\Delta(S)$ be the full simplex on S (so $\Delta(\varnothing) = \{\varnothing\}$).

Lemma A.2 ([Bjo95, Theorem 10.6], [Hat02, Corollary 4G.3])**.** *Let $L = \bigcup_{w \in W} \Delta(S_w)$ be a union of full simplices with nonempty vertex sets S_w, indexed by a finite set W. Then L is homotopy equivalent to its **witness nerve**, the complex on W whose faces are the nonempty sets $W' \subseteq W$ with $\bigcap_{w \in W'} S_w \neq \varnothing$.*

Lemma A.3. *Let M be a simplicial complex on a finite vertex set W, and let $t \geq 0$. Suppose that every subset of W with at most $t + 1$ elements is a face. Then $\widetilde{H}_s(M) = 0$ for $s < t$. Call a $(t+2)$-element subset of W* missing *if it is not a face. If for every missing set S there is $w \in W \smallsetminus S$ such that $S' \cup \{w\}$ is a face for every $(t+1)$-element subset $S' \subseteq S$, then also $\widetilde{H}_t(M) = 0$.*

Proof. The complex M and the full simplex on W have the same chain groups and boundary maps through degree t, which gives the first assertion. A t-cycle z of M bounds a $(t+1)$-chain $\sum_S c_S[S]$ in the full simplex. For a missing S replace $[S]$ by the cone $w * \partial[S]$, a chain of M by hypothesis, with $\partial(w * \partial[S]) = \partial[S]$. The resulting chain of M has boundary z. □

B The nine-point complex

In this section we prove that $\mathcal{C}_2(9,3)$ is acyclic through degree seven. For 3-sets A, Z one has $d_J(A,Z) \le 2$ if and only if $A \cap Z \neq \varnothing$. Hence a family of triples is a face of $\mathcal{C}_2(9,3)$ exactly when it has a *transversal* of three points, that is, a 3-set meeting each of its members. We abbreviate sets by strings of digits, writing 123 for $\{1,2,3\}$.

For a family R of subsets of [8] let $\tau(R)$ be the least size of a transversal of R contained in [8], and let $W(R)$ be the set of 3-subsets of [8] which are transversals of R. For a set W of triples in [8] let $\mathcal{N}_2(W)$ be the complex on W whose faces are the subfamilies having a transversal of at most two points of [8].

B.1 Triples on eight points

Let

$$\mathcal{X} = \{123,\ 124,\ 345,\ 678\}.$$

Its incidences are shown in Figure 8: the first three triples pairwise intersect, in 12, 3 and 4, and the fourth is disjoint from them. A transversal of $\mathcal{X}$ must meet 678, and no point lies in all of $123, 124, 345$; so $\tau(\mathcal{X}) = 3$.

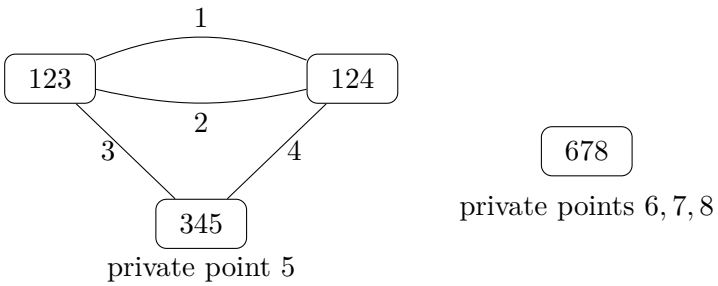


Figure 8: The family $\mathcal{X}$. An edge records a point lying in the two triples it joins.

Lemma B.1 (Triples on eight points)**.** *Let R be a family of distinct triples in* [8].

(a) If $|R| \le 3$ then $\tau(R) \le 2$.

(b) If $|R| = 4$ and $\tau(R) = 3$, then R is the image of $\mathcal{X}$ under a permutation of [8].

(c) The triples $B \notin \mathcal{X}$ for which some three members of $\mathcal{X}$ together with B have no transversal of two points are exactly $125, 135, 145, 235, 245$.

(d) If $|R| \le 5$ then $|W(R)| \ge 10$.

Proof. (a) Three triples have nine incidences on eight points, so two of them share a point x; then x and a point of the third triple form a transversal.

(b) No point lies in three members of R, since such a point together with a point of the fourth member would be a transversal. Form a multigraph on the four members, with one edge for each point lying in two of them. If e points lie in two members and s points in one, then $2e + s = 12$ and $e + s \le 8$, so $e \ge 4$. Two edges without a common end would give a two-point transversal, so the edges pairwise share an end. Hence they all pass through one member, or all join members of a single triangle. A member has three points, so it carries at most three edges, and the first

alternative is impossible. In a triangle the degree sum is $2e \geq 8$, while each degree is at most 3; so $e = 4$, the multiplicities on the three sides are $2, 1, 1$ (a multiplicity 3 would make two members equal, and multiplicities $2, 2, 0$ would give a member of degree four), and the fourth member carries no edge. Thus three members have the shape of $123, 124, 345$, the fifth point of their union lies in only one of them, and the fourth member consists of the three remaining points.

(c) Let $\mathcal{X}'$ consist of three members of $\mathcal{X}$ such that $\mathcal{X}' \cup \{B\}$ has no two-point transversal. By (b) it has the shape of $\mathcal{X}$: three pairwise intersecting members with intersections of sizes $2, 1, 1$, no point in three of them, and a fourth member disjoint from these. If $\mathcal{X}' = \{123, 124, 345\}$, then B is the disjoint member, so $B = 678$. Otherwise $678 \in \mathcal{X}'$ is the disjoint member, and $B \subseteq 12345$ completes the other two members of $\mathcal{X}'$ to the intersecting part. If these are $123, 124$, then B meets each of them in one point and avoids $1, 2$, so $B = 345$. If they are $123, 345$, then B avoids 3 and either contains 12 and one of $4, 5$, or contains 45 and one of $1, 2$; this gives $124, 125, 145, 245$. If they are $124, 345$, one obtains in the same way $123, 125, 135, 235$. Removing the members of $\mathcal{X}$ leaves the five triples stated.

(d) Adding triples can only remove transversals, so we may assume that $R = \{R_1, \dots, R_5\}$. For a triple $Z \subseteq [8]$ let m_Z be the number of i with $R_i \cap Z = \varnothing$. Each R_i is disjoint from $\binom{5}{3} = 10$ triples, so $\sum_Z m_Z = 50$. Put $e = \sum_{m_Z > 0}(m_Z - 1)$; then exactly $50 - e$ triples have $m_Z > 0$, and $|W(R)| = 56 - 50 + e = 6 + e$. We show $e \geq 4$. This is clear if some $m_Z = 5$. Otherwise $\binom{m_Z}{2} \leq 2(m_Z - 1)$ whenever $m_Z > 0$, and

$$2e \geq \sum_Z \binom{m_Z}{2} = \sum_{i<j} \binom{2 + |R_i \cap R_j|}{3} \geq \sum_{i<j} |R_i \cap R_j|,$$

since $[8] \smallsetminus (R_i \cup R_j)$ has $2 + |R_i \cap R_j|$ points. The last sum equals $\sum_x \binom{\delta_x}{2}$, where δ_x is the number of members containing x; as $\sum_x \delta_x = 15$ over eight points and $\binom{\delta}{2} \geq \delta - 1$, it is at least 7. Hence $e \geq 4$. □

Lemma B.2 (Two-point transversals)**.** *Let W be a nonempty set of triples in* [8]*. Then $\widetilde{H}_0(\mathcal{N}_2(W)) = H_1(\mathcal{N}_2(W)) = 0$. If $|W| \geq 10$, then also $H_2(\mathcal{N}_2(W)) = 0$.*

Proof. By Lemma B.1(a), every subfamily with at most three members is a face, and Lemma A.3 with $t = 2$ gives the first assertion. A missing four-element subfamily is a copy of $\mathcal{X}$, by Lemma B.1(b). By Lemma B.1(c), at most nine elements of W (the four members and the five triples listed there) fail to be cone vertices for it, so for $|W| \geq 10$ a cone vertex exists, and Lemma A.3 gives $H_2(\mathcal{N}_2(W)) = 0$. □

B.2 The fibers

Apply Lemma A.1 to $K = \mathcal{C}_2(9, 3)$, with the triples in [8] as base vertices and the triples containing 9 as fiber vertices. We identify the fiber vertex $\{9\} \cup X$ with the pair $X \subseteq [8]$.

Lemma B.3 (Fibers over triple families)**.** *Let R be a face of $\mathcal{C}_2(9, 3)$ consisting of triples in* [8]*.*

(a) If $\tau(R) \leq 2$, then K_R is the full simplex on $\binom{[8]}{2}$.

(b) If $\tau(R) = 3$, then $W(R) \neq \varnothing$ and

$$K_R = \bigcup_{Z \in W(R)} \Delta(\{X : X \cap Z \neq \varnothing\}) \simeq \mathcal{N}_2(W(R)).$$

Proof. (a) If $T \subseteq [8]$ is a transversal of R with $|T| \le 2$, then $T \cup \{9\}$ meets R and all fiber vertices. (b) A transversal of R together with some fiber vertices, having at most three points, cannot contain 9, since 9 lies in no member of R and removing it would leave a transversal of R with at most two points. So it is a member of $W(R)$ meeting each of the pairs X. This proves the displayed equality; $W(R) \neq \varnothing$ because R is a face. A set $W' \subseteq W(R)$ has a common vertex X in the simplices exactly when a set of at most two points meets every member of W', so the witness nerve is $\mathcal{N}_2(W(R))$, and Lemma A.2 applies. □

B.3 The exceptional fiber

The degree budget of Lemma A.1 is most demanding for the family $\mathcal{X}$, where degrees up to three are required.

Lemma B.4 (The exceptional fiber)**.** $\widetilde{H}_t(K_{\mathcal{X}}) = 0$ *for* $-1 \le t \le 3$.

Proof. Put $A = 12345$, $J = 678$, and let Γ be the set of pairs in A meeting each of $123, 124, 345$:

$$\Gamma = \{13,\ 14,\ 15,\ 23,\ 24,\ 25,\ 34\}.$$

We regard Γ as a graph on A; every point of A lies on an edge of Γ.

Step 1 (the centers). A member of $W(\mathcal{X})$ meets 678. It cannot contain two points of J, since no point lies in all of $123, 124, 345$. Hence

$$W(\mathcal{X}) = \{P \cup \{e\} :\ P \in \Gamma,\ e \in J\},$$

twenty-one triples; we call e the *type* of the center $P \cup \{e\}$.

Step 2 (three cones). For $e \in J$ let $K_e \subseteq K_{\mathcal{X}}$ consist of the families of pairs met by a single center of type e. Then $K_{\mathcal{X}} = K_6 \cup K_7 \cup K_8$. Any pair containing e meets every center of type e; hence each K_e, and each $K_e \cap K_{e'}$ $(e \neq e')$, is a cone with apex $\{e, e'\}$. Put $\Theta = K_6 \cap K_7 \cap K_8$. The Mayer–Vietoris sequence for augmented chains shows, in turn, that $K_6 \cup K_7$ is acyclic; that $\widetilde{H}_t((K_6 \cup K_7) \cap K_8) \cong \widetilde{H}_{t-1}(\Theta)$, because $(K_6 \cup K_7) \cap K_8 = (K_6 \cap K_8) \cup (K_7 \cap K_8)$ is a union of two cones meeting in Θ; and finally that

$$\widetilde{H}_t(K_{\mathcal{X}}) \cong \widetilde{H}_{t-2}(\Theta) \qquad \text{for all } t.$$

It therefore suffices to show that Θ is nonempty and connected and that $H_1(\Theta) = 0$.

Step 3 (the vertices of Θ). A family $\mathcal{S}$ of pairs lies in Θ exactly when for each $e \in J$ some $P_e \in \Gamma$ satisfies: $P_e \cup \{e\}$ meets every member of $\mathcal{S}$. A pair inside J misses every center of the third type. Hence the vertices of Θ are the ten pairs inside A, which we call *inner*, and the fifteen pairs $\{x, e\}$ with $x \in A$, $e \in J$, which we call *cross* vertices.

Step 4 (the inner part). Let Θ_A be the subcomplex induced on the inner vertices. A family of inner pairs lies in Θ exactly when a single edge $P \in \Gamma$ meets all of them (use P for all three types; conversely the witness for type 6 works). So $\Theta_A = \bigcup_{P \in \Gamma} \Delta(\{\text{inner pairs meeting } P\})$, and by Lemma A.2 it is homotopy equivalent to the complex on Γ whose faces are the sets of edges having a transversal of two points of A. Three edges always have one: three disjoint pairs would need six points, so two of them share a point, which together with a point of the third edge is a transversal. By Lemma A.3, Θ_A is connected and $H_1(\Theta_A) = 0$.

Step 5 (the cross vertices). Let $v = \{x, e\}$ be a cross vertex. Three observations hold.

(i) The inner pairs containing x, together with v, form a face of Θ: use an edge of Γ through x for all three types.

(ii) If an inner vertex Y is adjacent to v, there is an edge $P \in \Gamma$ through x with $\{v, Y, P\} \in \Theta$. Indeed, choose $e' \in J \smallsetminus \{e\}$; the witness $P_{e'} \cup \{e'\}$ for $\{v, Y\}$ must meet v in x, so $P = P_{e'}$ contains x and meets Y, and using P for all three types shows $\{v, Y, P\} \in \Theta$.

(iii) If $v' = \{x', e'\}$ is a cross vertex adjacent to v, there is an inner vertex c with $\{v, v', c\} \in \Theta$. If $x = x'$ take for c any edge of Γ through x. If $x \neq x'$, choose $e'' \in J \smallsetminus \{e, e'\}$; the witness of type e'' for $\{v, v'\}$ must contain x and x', so $\{x, x'\} \in \Gamma$, and $c = \{x, x'\}$ works with this edge used for all three types.

Step 6 (conclusion). By (i) every cross vertex is adjacent to an inner vertex, and Θ_A is connected; so Θ is nonempty and connected. Let γ be a closed edge path in Θ. Replace each edge vv' between cross vertices by the path vcv' of (iii); the difference is the boundary of the triangle $\{v, v', c\}$. Each remaining visit to a cross vertex v has the form YvY' with Y, Y' inner. Consider the graph on the inner neighbors of v in which Y_1, Y_2 are joined when $\{v, Y_1, Y_2\} \in \Theta$. By (i) the inner pairs through x form a complete subgraph, and by (ii) every vertex is joined to one of them; so this graph is connected, and contains a path $Y = Y_0, Y_1, \dots, Y_q = Y'$. The triangles $\{v, Y_l, Y_{l+1}\}$ show that YvY' is homologous, relative to its ends, to the path $Y_0Y_1 \cdots Y_q$ in Θ_A. After these replacements γ is homologous to a cycle of Θ_A, which bounds there. Since every integral 1-cycle is a sum of closed edge paths, $H_1(\Theta) = 0$. □

Proposition B.5 (The nine-point vanishing)**.** $\widetilde{H}_j(\mathcal{C}_2(9,3); \mathbb{Z}) = 0$ *for* $0 \leq j \leq 7$.

Proof. By Lemma A.1 with $d = 7$ it suffices to show $\widetilde{H}_t(K_R) = 0$ for $-1 \leq t \leq 7 - |R|$, for every face R of triples in [8]. If $\tau(R) \leq 2$, which includes all R with $|R| \leq 3$ by Lemma B.1(a), the fiber is a full simplex. Let $\tau(R) = 3$; then $K_R \simeq \mathcal{N}_2(W(R))$ with $W(R) \neq \varnothing$, and $|R| \geq 4$. If $|R| \geq 6$, only $t \leq 1$ is required, and Lemma B.2 applies. If $|R| = 5$, then $t \leq 2$ is required, and Lemma B.1(d) and Lemma B.2 apply. If $|R| = 4$, then R is a copy of $\mathcal{X}$ by Lemma B.1(b), and Lemma B.4 supplies all degrees $t \leq 3$. □

C The eight-point complex

In this section we prove that $\mathcal{C}_2(8,4)$ is acyclic through degree seven. We identify [8] with $\Omega = \mathbb{F}_2^3$; the labels are the seventy 4-subsets of Ω. The fourteen affine planes play a special role: they will be the fiber vertices in Lemma A.1, and a Fourier transform places them on the fourteen signed coordinate axes of $\mathbb{R}^7$.

The uniform fiber bound handles families with at least five nonplane labels. For smaller families, a degree shift leaves three borderline configurations, treated by the majority, one-missing-plane, and ten-center lemmas. The table in the proof of Proposition C.8 records how these inputs fit together.

C.1 Signed vectors

For $u, x \in \Omega$ let $u \cdot x \in \mathbb{F}_2$ be the dot product and $\chi_u(x) = (-1)^{u \cdot x}$. For a 4-set A define $v_A \in \mathbb{R}^7$, with coordinates indexed by $u \in \Omega \smallsetminus \{0\}$, by

$$v_A(u) = \tfrac{1}{2} \sum_{x \in A} \chi_u(x).$$

Let $\mathcal{P}$ be the set of affine planes (cosets of 2-dimensional subspaces); the other 56 labels are *nonplanes*. Put $s(A) = \sum_{x\in A} x \in \Omega$, and for $a \neq 0$ let $S_a = \{u : \ u \cdot a = 1\}$, a set of four coordinates. We write e_u for the standard basis vectors of $\mathbb{R}^7$ and $A^c = \Omega \smallsetminus A$.

Lemma C.1 (The Fourier coordinates)**.** *(a) $v_A \cdot v_Z = 2|A \cap Z| - 4$. Hence $\|v_A\|^2 = 4$, $v_{A^c} = -v_A$, the map $A \mapsto v_A$ is injective, and Z is a center of A if and only if $v_A \cdot v_Z \geq 0$. If $v_{Z_1} + \cdots + v_{Z_t} = 0$ then t is even.*

(b) A is a plane if and only if $s(A) = 0$. The plane $\{x : \ u \cdot x = b\}$ has the vector $2(-1)^b e_u$, so the planes correspond to the fourteen vectors $\pm 2e_u$.

(c) If A is a nonplane and $a = s(A)$, then v_A is supported on S_a, with entries ± 1 whose product is -1. Each of the $7 \cdot 8 = 56$ sign patterns of this kind is v_A for exactly one nonplane A.

(d) Distinct supports $S_a \neq S_b$ meet in two coordinates. Two sign patterns on the same support, each with product -1, agree in an even number of coordinates.

Proof. (a) For $g\colon \Omega \to \mathbb{R}$ put $\hat{g}(u) = \sum_x g(x)\chi_u(x)$. Orthogonality of characters gives $\sum_{u\in\Omega} \hat{g}(u)\hat{h}(u) = 8\sum_x g(x)h(x)$ and the inversion formula $g = \frac{1}{8}\sum_u \hat{g}(u)\chi_u$. For $g_A = \mathbf{1}_A - \frac{1}{2}$ one has $\hat{g}_A(0) = 0$ and $\hat{g}_A(u) = 2v_A(u)$ for $u \neq 0$; hence $4\, v_A \cdot v_Z = 8\sum_x g_A g_Z = 8(|A \cap Z| - 2)$. Also $g_{A^c} = -g_A$. By inversion v_A determines g_A, and if $\sum_i v_{Z_i} = 0$ then $\sum_i g_{Z_i} = 0$, so that $\sum_i \mathbf{1}_{Z_i}$ is the constant $t/2$, which must be an integer.

(b), (c) The product of the four signs $\chi_u(x)$, $x \in A$, equals $\chi_u(s(A))$. If $u \cdot s(A) = 1$, an odd number of them is -1 and $v_A(u) = \pm 1$; if $u \cdot s(A) = 0$, an even number is -1 and $v_A(u) \in \{0, \pm 2\}$. If $a = s(A) \neq 0$, the four coordinates of S_a already contribute $4 = \|v_A\|^2$, so all other entries vanish. If $s(A) = 0$, all entries lie in $\{0, \pm 2\}$, so exactly one is nonzero, say $v_A = 2(-1)^b e_u$; then $\chi_u(x) = (-1)^b$ for all $x \in A$, and A is the plane $\{x : u \cdot x = b\}$. Conversely the four points $p, p+v, p+w, p+v+w$ of a plane sum to 0. This proves (b) and the first part of (c).

Translating A by y multiplies $v_A(u)$ by $\chi_u(y)$ and does not change $s(A)$; the product of these factors over S_a is $\chi_{\sum_{u\in S_a} u}(y) = 1$, because S_a is an affine plane in the space of characters and its four elements sum to 0. An invertible linear map g of Ω satisfies $v_{gA}(u) = v_A(g^\top u)$, and so permutes coordinates. Nonplanes are exactly the affinely independent 4-sets (an affine dependence among four distinct points of Ω says that their sum is 0), and the affine group acts transitively on them; so the sign product is the same for all nonplanes. For $A = \{0, e_1, e_2, e_3\}$ one finds $a = (1,1,1)$, $S_a = \{100, 010, 001, 111\}$, and entries $1, 1, 1, -1$; the product is -1. Finally there are $\binom{8}{4} - 14 = 56$ nonplanes, $7 \cdot 8$ such sign patterns, and $A \mapsto v_A$ is injective.

(d) The conditions $u \cdot a = u \cdot b = 1$ are independent, so they have two solutions. On a common support, the coordinatewise product of two patterns of product -1 has product $+1$; so they differ, and hence agree, in an even number of coordinates. □

C.2 Exclusions

For a label Z, its **exclusion** is the set of planes for which Z is not a center:

$$\mathrm{Ex}(Z) = \{p \in \mathcal{P} : \ v_p \cdot v_Z < 0\}.$$

By Lemma C.1, the plane with vector $2\eta e_u$ ($\eta = \pm 1$) lies in $\mathrm{Ex}(Z)$ exactly when $\eta\, v_Z(u) < 0$. Thus $\mathrm{Ex}(p) = \{p^c\}$ for a plane p, while a nonplane excludes four planes, one on each coordinate of its support, of sign opposite to the entry there; the exclusion of a nonplane determines its vector. For

a set $\mathcal{S}$ of labels put $\mathrm{Ex}(\mathcal{S}) = \bigcup_{Z\in\mathcal{S}} \mathrm{Ex}(Z)$, and say that $\mathcal{S}$ *covers* $\mathcal{P}$ if $\mathrm{Ex}(\mathcal{S}) = \mathcal{P}$. The following reformulation is used throughout:

$\mathcal{S}$ covers $\mathcal{P}$ if and only if, at every coordinate u, the entries $v_Z(u)$, $Z \in \mathcal{S}$, include a positive and a negative number. (C.1)

Lemma C.2 (Three and four exclusions)**.** *(a) For any three labels,* $|\mathrm{Ex}(Z_1)\cup\mathrm{Ex}(Z_2)\cup\mathrm{Ex}(Z_3)| \le 11$. *In particular three labels never cover* $\mathcal{P}$.

(b) If four labels $Z_1, \dots, Z_4$ *cover* $\mathcal{P}$*, they are nonplanes in* ***star position****: there are seven distinct coordinates* u_0 *and* $u_{ij} = u_{ji}$ *(*$1 \le i < j \le 4$*) such that the support of* v_{Z_i} *is* $\{u_0\}\cup\{u_{ij} : j \ne i\}$*, the two nonzero entries at each* u_{ij} *are opposite, and the four entries at* u_0 *are two* $+1$*'s and two* -1*'s. In particular* $\sum_i v_{Z_i} = 0$.

Proof. (a) By (C.1), covering requires two nonzero entries at each of the seven coordinates, whereas three labels have at most twelve nonzero entries. For the sharper bound, a union of size twelve requires three nonplanes with pairwise disjoint exclusions. Two exclusions are disjoint exactly when the two vectors have opposite signs at every common coordinate. If two of the three vectors have the same support, they are therefore opposite; the third vector meets this support in at least two coordinates, and there it agrees in sign with one of them. If the three supports are distinct, then $v_{Z_i} \cdot v_{Z_j} = -2$ for $i \ne j$ (two common coordinates, opposite signs), and $\|v_{Z_1} + v_{Z_2} + v_{Z_3}\|^2 = 12 - 12 = 0$, contrary to Lemma C.1(a).

(b) If one label is a plane, the supports have total size at most $1 + 12 < 14$, which is impossible by (C.1). Let $a_i = s(Z_i)$ and $\Phi(u) = (u \cdot a_1, \dots, u \cdot a_4) \in \mathbb{F}_2^4$. The map Φ is linear, and u lies in the support of v_{Z_i} exactly when $\Phi(u)_i = 1$. By (C.1), $\Phi(u)$ has weight at least two for every $u \ne 0$. So Φ is injective, and its image is a 3-dimensional subspace, the kernel of $t \mapsto \sum_{i\in I} t_i$ for a nonempty $I \subseteq [4]$. This kernel contains the unit vectors f_i, $i \notin I$, of weight one; hence $I = [4]$, and the image consists of the vectors of even weight: the six vectors $f_i + f_j$ and $(1,1,1,1)$. Put $u_{ij} = \Phi^{-1}(f_i + f_j)$ and $u_0 = \Phi^{-1}(1,1,1,1)$. By (C.1) the two entries at u_{ij} are opposite. The sixteen nonzero entries have product $(-1)^4 = 1$ by Lemma C.1(c), and the twelve entries at the u_{ij} have product $(-1)^6 = 1$; so the four entries at u_0 have product 1, and since both signs occur there, two are positive and two negative. □

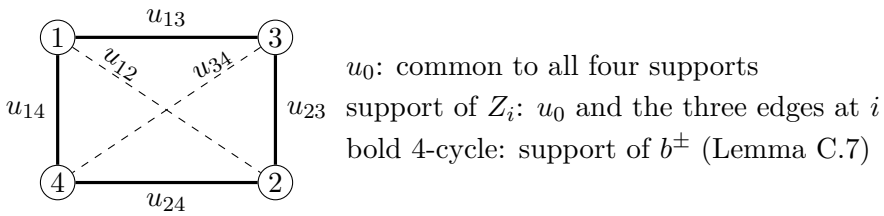


Figure 9: Star position. The coordinates u_{ij} are the edges of the complete graph on $\{1,2,3,4\}$; u_0 is the remaining coordinate.

C.3 The fibers and a uniform bound

Apply Lemma A.1 to $K = \mathcal{C}_2(8,4)$, with the nonplanes as base vertices and the planes as fiber vertices. For a face R of nonplanes let $W(R)$ be its set of centers, which is nonempty. A set of

planes is a face of K_R exactly when some center of R excludes none of them, so

$$K_R = \bigcup_{Z \in W(R)} \Delta(\mathcal{P} \smallsetminus \mathrm{Ex}(Z)).$$

By Lemma A.2, K_R is homotopy equivalent to the witness nerve $\mathcal{N}_R$, the complex on $W(R)$ whose faces are the sets of centers which do not cover $\mathcal{P}$.

Lemma C.3 (A uniform fiber bound)**.** *For every face R of nonplanes ($R = \varnothing$ allowed), $\widetilde{H}_t(K_R) = 0$ for $-1 \le t \le 2$.*

Proof. By Lemma C.2(a), every set of at most three centers is a face of $\mathcal{N}_R$, so $\widetilde{H}_t(\mathcal{N}_R) = 0$ for $t \le 1$ by Lemma A.3. Let $\{Z_1, \dots, Z_4\} \subseteq W(R)$ be missing. By Lemma C.2(b) the Z_i are in star position and $\sum_i v_{Z_i} = 0$. For $A \in R$ the four numbers $v_A \cdot v_{Z_i}$ are nonnegative with sum 0, so they vanish; hence $v_A \cdot v_{Z_1^c} = -v_A \cdot v_{Z_1} = 0$, and $Z_1^c \in W(R)$. It differs from $Z_1, \dots, Z_4$, since its support is that of Z_1 while its vector is $-v_{Z_1}$, and it is a cone vertex. Indeed, $\{Z_1^c, Z_2, Z_3, Z_4\}$ does not cover $\mathcal{P}$, since at u_{12} the only nonzero entries are those of Z_2 and Z_1^c, and these have the same sign. For $j \ne 1$ choose $k \notin \{1, j\}$; then $\{Z_1^c\} \cup (\{Z_1, \dots, Z_4\} \smallsetminus \{Z_j\})$ does not cover $\mathcal{P}$, since at u_{jk} only Z_k has a nonzero entry. Lemma A.3 gives $H_2(\mathcal{N}_R) = 0$. □

C.4 The degree shift

For the faces with at most four labels a finer analysis is needed. The point is that the plane centers of R contribute a contractible part, which shifts degrees.

Let R be a nonempty face of nonplanes, and put $m = |\mathrm{Ex}(R)|$, so that $4 \le m \le 14$. A plane is a center of R exactly when it lies outside $\mathrm{Ex}(R)$. Let

$$\mathcal{P}_0 = \{p^c : \ p \in \mathcal{P} \smallsetminus \mathrm{Ex}(R)\}, \qquad \mathcal{P}_1 = \mathcal{P} \smallsetminus \mathcal{P}_0,$$

so that $\mathcal{P}_0$ consists of the exclusions of the plane centers and $|\mathcal{P}_1| = m$. Call a nonplane center Z of R *essential* if $\mathrm{Ex}(Z) \subseteq \mathcal{P}_1$, and put

$$J_R = \bigcup_{Z \text{ essential}} \Delta(\mathcal{P}_1 \smallsetminus \mathrm{Ex}(Z)),$$

a complex on $\mathcal{P}_1$, possibly equal to $\{\varnothing\}$.

Lemma C.4 (The degree shift)**.** *Let R be a nonempty face of nonplanes, with $m = |\mathrm{Ex}(R)|$.*

(a) If $m < 14$ and R has no essential center, K_R is contractible.

(b) Otherwise $\widetilde{H}_t(K_R) \cong \widetilde{H}_{t+m-14}(J_R)$ for every t.

(c) Suppose that R has an essential center. Then J_R has a vertex if $m \ge 5$. Moreover, for $m \ge 5$, J_R is homotopy equivalent to the complex $\mathcal{N}'_R$ on the essential centers whose faces are the sets $\mathcal{S}$ with $\mathrm{Ex}(\mathcal{S}) \ne \mathcal{P}_1$; and J_R is connected if $m \ge 9$, and satisfies $H_1(J_R) = 0$ if $m \ge 12$.

Proof. A plane center p contributes the simplex $\Delta(\mathcal{P} \smallsetminus \{p^c\})$, and the union of these is

$$D_R = \{S \subseteq \mathcal{P} : \ S \not\supseteq \mathcal{P}_0\},$$

the join of the boundary of $\Delta(\mathcal{P}_0)$ with $\Delta(\mathcal{P}_1)$. If a nonplane center Z excludes some $p^c \in \mathcal{P}_0$, its simplex lies in $\Delta(\mathcal{P} \smallsetminus \{p^c\}) \subseteq D_R$. The simplex of an essential center consists of the sets $S_0 \cup S_1$ with $S_0 \subseteq \mathcal{P}_0$ and $S_1 \in \Delta(\mathcal{P}_1 \smallsetminus \mathrm{Ex}(Z))$. Hence

$$K_R = D_R \cup \{S_0 \cup S_1 : \ S_0 \subseteq \mathcal{P}_0, \ S_1 \in J_R\}.$$

(a) If $m < 14$ there is a plane center, so $\mathcal{P}_0 \neq \varnothing$, and $\mathcal{P}_1 \neq \varnothing$ because $m \geq 4$. Without essential centers $K_R = D_R$, which is a cone with any vertex of $\mathcal{P}_1$ as apex.

(b) If $m = 14$ there are no plane centers, every nonplane center is essential, and $K_R = J_R$. If $m < 14$, then D_R is contractible, so $\widetilde{H}_t(K_R) \cong H_t(K_R, D_R)$. The faces of K_R not in D_R are the sets $\mathcal{P}_0 \cup S$ with $S \in J_R$ (including $S = \varnothing$). Ordering $\mathcal{P}_0$ first, the boundary of $[\mathcal{P}_0] * [S]$ modulo D_R is $(-1)^{|\mathcal{P}_0|}[\mathcal{P}_0] * \partial[S]$, because deleting a vertex of $\mathcal{P}_0$ gives a face of D_R. So the relative chain complex is the augmented chain complex of J_R shifted up by $|\mathcal{P}_0| = 14 - m$.

(c) An exclusion of a nonplane has four elements, so for $m \geq 5$ every essential center contributes a nonempty simplex, and Lemma A.2 identifies J_R with $\mathcal{N}'_R$ up to homotopy. Two exclusions have at most eight elements, and three have at most eleven by Lemma C.2(a). So for $m \geq 9$ every pair of essential centers is a face of $\mathcal{N}'_R$, and for $m \geq 12$ every triple is; now apply Lemma A.3. □

C.5 Three borderline configurations

Three configurations are not settled by part (c) of Lemma C.4; the following lemmas treat them.

Lemma C.5 (Majority center)**.** *Let Z_1, Z_2, Z_3 be labels with $|\operatorname{Ex}(Z_1) \cup \operatorname{Ex}(Z_2) \cup \operatorname{Ex}(Z_3)| = 11$, and let M be the set of points lying in at least two of them. Then M is a 4-set, and*

(1) M is a center of every label of which Z_1, Z_2, Z_3 are centers;

(2) $\operatorname{Ex}(M) \subseteq \operatorname{Ex}(Z_1) \cup \operatorname{Ex}(Z_2) \cup \operatorname{Ex}(Z_3)$;

(3) for each pair i, j, the set $\operatorname{Ex}(M) \cup \operatorname{Ex}(Z_i) \cup \operatorname{Ex}(Z_j)$ is strictly smaller than $\operatorname{Ex}(Z_1) \cup \operatorname{Ex}(Z_2) \cup \operatorname{Ex}(Z_3)$.

Proof. Write $E_i = \operatorname{Ex}(Z_i)$, $v_i = v_{Z_i}$, and $E = E_1 \cup E_2 \cup E_3$. If some Z_i were a plane, $|E| \leq 9$; so all three are nonplanes. As $|E_1| + |E_2| + |E_3| = 12$, exactly one plane is excluded twice; relabel so that $|E_1 \cap E_2| = 1$ and E_3 is disjoint from $E_1 \cup E_2$. Two exclusions share the plane on a coordinate u exactly when the two vectors have nonzero entries of the same sign at u. Thus v_1 and v_2 agree in sign at exactly one common coordinate; by Lemma C.1(d) their supports are distinct, they have two common coordinates, and $v_1 \cdot v_2 = 0$. The vector v_3 is opposite to v_1 and to v_2 at all common coordinates. If v_3 had the support of v_1, then $v_3 = -v_1$, and at the two coordinates common to v_1 and v_2 the vector v_3 would agree in sign with v_2 once; the same argument applies to the support of v_2. So the three supports are distinct, and $v_1 \cdot v_3 = v_2 \cdot v_3 = -2$. By Lemma C.1(a), $|Z_1 \cap Z_2| = 2$ and $|Z_1 \cap Z_3| = |Z_2 \cap Z_3| = 1$, and inclusion–exclusion gives $|Z_1 \cup Z_2 \cup Z_3| = 8 + |Z_1 \cap Z_2 \cap Z_3| \leq 8$. Hence no point lies in all three sets, their union is Ω, $|M| = 4$, and $\mathbf{1}_M = \mathbf{1}_{Z_1} + \mathbf{1}_{Z_2} + \mathbf{1}_{Z_3} - \mathbf{1}$. Since $\mathbf{1}_M - \frac{1}{2} = \sum_i (\mathbf{1}_{Z_i} - \frac{1}{2})$, this gives

$$v_M = v_1 + v_2 + v_3.$$

(1) If $v_A \cdot v_i \geq 0$ for $i = 1, 2, 3$, then $v_A \cdot v_M \geq 0$. (2) If $v_p \cdot v_M < 0$, then $v_p \cdot v_i < 0$ for some i. (3) One has $v_M \cdot v_1 = v_M \cdot v_2 = 2$ and $v_M \cdot v_3 = 0$. We claim that $|\operatorname{Ex}(M) \cap E_i| \leq 2$ for $i = 1, 2$ and $E_3 \not\subseteq \operatorname{Ex}(M)$. If M is a plane, $\operatorname{Ex}(M)$ has one element. If M is a nonplane, it cannot have the support of v_i $(i = 1, 2)$, since an inner product 2 on a common support of four coordinates would mean three agreements, contrary to Lemma C.1(d); so the supports meet in two coordinates and $|\operatorname{Ex}(M) \cap E_i| \leq 2$. If $E_3 \subseteq \operatorname{Ex}(M)$, then the two exclusions are equal, so $v_M = v_3$ and $v_1 + v_2 = 0$, contrary to $v_1 \cdot v_2 = 0$. Now $\operatorname{Ex}(M) \cup E_1 \cup E_2$ misses a plane of E_3, and $\operatorname{Ex}(M) \cup E_1 \cup E_3$ misses one of the three planes of $E_2 \smallsetminus E_1$; the case $\operatorname{Ex}(M) \cup E_2 \cup E_3$ is symmetric. □

Lemma C.6 (One missing plane)**.** *(a) Let $\mathcal{A} = \{A_1, \ldots, A_4\}$ be nonplanes with $\operatorname{Ex}(\mathcal{A}) = \mathcal{P} \smallsetminus \{q\}$, where q has the vector $2e_h$, so that $q = \{x : h \cdot x = 0\}$. Put $d = \sum_i v_{A_i}$ and $w = \sum_i \mathbf{1}_{A_i} - 2 \cdot \mathbf{1}$, a function on Ω. Then one of the following holds:*

(i) $d = 4e_h$;

(ii) $d = 2e_h + 2\delta e_u$ *for a coordinate* $u \neq h$ *and* $\delta = \pm 1$;

(iii) the vectors v_{A_i} *are linearly independent, and* $w = \delta_x - \delta_y$ *with* $x \in q$, $y \notin q$, *where* δ_x *is the indicator function of the point* x.

(b) (Opposite holes.) *Let* $\mathcal{A}$ *and* $\mathcal{A}'$ *be sets of four nonplanes with* $\mathrm{Ex}(\mathcal{A}) = \mathcal{P} \smallsetminus \{q\}$ *and* $\mathrm{Ex}(\mathcal{A}') = \mathcal{P} \smallsetminus \{q^c\}$ *for some plane* q. *Then* $v_A \cdot v_{A'} < 0$ *for some* $A \in \mathcal{A}$ *and* $A' \in \mathcal{A}'$.

Proof. (a) Since q is not excluded, all entries at h are nonnegative, and since q^c is excluded, one of them is positive. Every other coordinate carries entries of both signs. Define Φ as in the proof of Lemma C.2(b). Now $\Phi(u)$ has weight at least two for $u \notin \{0, h\}$ and at least one for $u = h$; so Φ is injective, and its image is the kernel of $t \mapsto \sum_{i \in I} t_i$ for some nonempty I. The image contains $4 - |I|$ unit vectors, and at most one vector of weight one is allowed; so $|I| \geq 3$.

If $|I| = 4$, the image consists of the vectors of even weight, and the supports have the star shape of Lemma C.2(b), with coordinates u_0 and u_{ij}; at each $u_{ij} \neq h$ the two entries are opposite. If $h = u_0$, all four entries at h are $+1$, and $d = 4e_h$. If $h = u_{ij}$, both entries at h are $+1$, so the twelve entries at the pair coordinates have product $(-1)^5 = -1$. The sixteen entries have product 1, so the four entries at u_0 have product -1 and sum ± 2. This gives (ii) with $u = u_0$.

If $|I| = 3$, say $I = \{1, 2, 3\}$, the nonzero vectors of the image are f_4, which must be $\Phi(h)$, the three vectors $f_i + f_j$ $(i < j \leq 3)$, and the three vectors $f_i + f_j + f_4$. At h only v_{A_4} is nonzero, with entry 1; at the coordinates of weight two the entries are opposite; at those of weight three both signs occur among three entries ± 1. Hence d has four entries ± 1, with $d(h) = 1$, and $\|d\|^2 = 4$. The function w has sum 0 and transform $\hat{w}(u) = 2d(u)$ for $u \neq 0$; so $8 \sum_x w(x)^2 = \sum_u \hat{w}(u)^2 = 16$. An integer-valued function with sum 0 and $\sum w^2 = 2$ has the form $\delta_x - \delta_y$, and $2 = \hat{w}(h) = \chi_h(x) - \chi_h(y)$ gives $x \in q$, $y \notin q$. Finally, in a relation $\sum_i c_i v_{A_i} = 0$ the coordinate h gives $c_4 = 0$, the coordinates $f_i + f_j$ give $c_1 = c_2 = c_3$, and then $c_1 = 0$ because three vectors v_A never sum to zero (Lemma C.1(a)).

(b) Complementing all sets negates all vectors, preserves inner products, and replaces each excluded plane by its complement; so we may assume that q has the vector $2e_h$. Suppose that all sixteen inner products are nonnegative. Every $v' = v_{A'}$, $A' \in \mathcal{A}'$, has $v'(h) \leq 0$, and one of them is negative. If $\mathcal{A}$ is of type (i), then $4v'(h) = d \cdot v' \geq 0$, so $v'(h) = 0$ for all A', a contradiction. If $\mathcal{A}$ is of type (ii), then $d \cdot v' \geq 0$ gives $\delta v'(u) \geq -v'(h) \geq 0$ for all A', so the plane with vector $2\delta e_u$ is not excluded by $\mathcal{A}'$; as $u \neq h$ it differs from q^c, a contradiction. The pair formed by the complements of the members of $\mathcal{A}'$ and those of $\mathcal{A}$ satisfies the same hypotheses, with the same q; so the same argument shows that the family of complements of the members of $\mathcal{A}'$ is also of type (iii). Complementation replaces $w' = \sum \mathbf{1}_{A'} - 2 \cdot \mathbf{1}$ by $-w'$. Hence $w = \delta_x - \delta_y$ with $x \in q$, $y \notin q$, and $w' = \delta_{x'} - \delta_{y'}$ with $x' \notin q$, $y' \in q$. Then $x \neq x'$ and $y \neq y'$, and

$$\sum_{A,A'} v_A \cdot v_{A'} = \tfrac{1}{4} \sum_{u \neq 0} \hat{w}(u)\hat{w}'(u) = 2 \sum_x w(x)w'(x) = -2([x = y'] + [y = x']) \leq 0.$$

So all sixteen inner products vanish, and the two families span orthogonal 4-dimensional subspaces of $\mathbb{R}^7$, which is impossible. □

Lemma C.7 (Ten centers)**.** *Let* $R = \{A_1, \ldots, A_4\}$ *consist of nonplanes which cover* $\mathcal{P}$. *Then* R *has exactly ten centers, and there are two disjoint four-element sets* F_+, F_- *of centers such that a set of at most five centers covers* $\mathcal{P}$ *if and only if it contains* F_+ *or* F_-.

Proof. By Lemma C.2(b) the vectors $v_i = v_{A_i}$ are in star position. Put $s_i = v_i(u_0)$ and $\varepsilon_{ij} = v_i(u_{ij})$, so that $\varepsilon_{ji} = -\varepsilon_{ij}$, and relabel so that $(s_1, s_2, s_3, s_4) = (1, 1, -1, -1)$. Lemma C.1(c) gives

$$s_i \prod_{j \neq i} \varepsilon_{ij} = -1 \qquad (i = 1, \ldots, 4). \tag{C.2}$$

Step 1 (the centers). For a center Z the four numbers $v_Z \cdot v_i$ are nonnegative with sum $v_Z \cdot 0 = 0$, so v_Z is orthogonal to every v_i. A plane center would need a coordinate outside all four supports, so the centers are nonplanes. Under Φ the coordinates correspond to the nonzero vectors of even weight in $\mathbb{F}_2^4$, and the seven supports to the seven nonzero linear forms on that space: the forms t_i give the four star supports $S_i = \{u_0\} \cup \{u_{ij} : j \neq i\}$, and the forms $t_i + t_j = t_k + t_l$ give the three 4-cycles $\{u_{ik}, u_{il}, u_{jk}, u_{jl}\}$ of the complete graph, one for each splitting $\{i,j\} \cup \{k,l\}$ of [4] (Figure 9).

Star supports. Let Z have support S_i, with entry t at u_0 and y_j at u_{ij}. Orthogonality to v_j ($j \neq i$), whose support meets S_i in $\{u_0, u_{ij}\}$, reads $ts_j + \varepsilon_{ji} y_j = 0$, that is, $y_j = ts_j\varepsilon_{ij}$. Orthogonality to v_i is then $t\sum_j s_j = 0$. The sign product is $t\prod_{j\neq i} ts_j\varepsilon_{ij} = s_i \cdot (-s_i) = -1$ by (C.2), since $\prod_{j\neq i} s_j = s_i$. So for each i and each $t = \pm 1$ there is exactly one center with support S_i and entry t at u_0; call it w_i^t.

Cycle supports. Let Z have the support $\{u_{ik}, u_{il}, u_{jk}, u_{jl}\}$, with entries $b(u_{xy})$, and put $g(x,y) = \varepsilon_{xy} b(u_{xy})$, so that $g(y,x) = -g(x,y)$. Orthogonality to v_i reads $g(i,k) + g(i,l) = 0$, and similarly at j,k,l: at every vertex of the 4-cycle $i\,k\,j\,l$ the two outgoing values cancel. Hence g is a unit circulation around the cycle, in one of its two directions; then $b(u_{ik})b(u_{il})b(u_{jk})b(u_{jl}) = \varepsilon_{ik}\varepsilon_{il}\varepsilon_{jk}\varepsilon_{jl}$. Multiplying (C.2) for i and j and using $\varepsilon_{ij}\varepsilon_{ji} = -1$ shows that this product equals $-s_i s_j$. A nonplane requires the value -1, that is, $s_i = s_j$. This happens only for the splitting $\{1,2\} \cup \{3,4\}$. So there are exactly two centers, b^+ and b^-, with support $\{u_{13}, u_{14}, u_{23}, u_{24}\}$, and none with the other two cycle supports. In all there are ten centers.

Step 2 (covers by at most five centers). Give w_i^t the *color* ts_i, and call $\{w_i^+, w_i^-\}$ *group* i. At u_{ij}, after multiplication by ε_{ij}, the entry of w_i^t is $ts_j = (ts_i)s_i s_j$, and that of $w_j^{t'}$ is $-t's_i = -(t's_j)s_i s_j$. Hence, at u_{ij}, a center of group i and one of group j supply both signs exactly when they have the same color, while the two centers of one group always do. The centers $b^\pm$ vanish at u_0, u_{12} and u_{34}. By (C.1), a set covers $\mathcal{P}$ when both signs occur at every coordinate.

Let $\mathcal{S}$ be a set of at most five centers covering $\mathcal{P}$. Coverage at u_{12} needs two centers from groups $1,2$, and at u_{34} two from groups $3,4$; so $\mathcal{S}$ contains at least four w's and at most one b. Every group occurs. Indeed, suppose group i is absent; let i' be its partner in $\{1,2\}$ or $\{3,4\}$, and $\{j,k\}$ the other pair. Coverage at $u_{ii'}$ forces both centers of group i'. If group j were also absent, only a b could contribute at u_{ij}, giving at most one sign. So groups j and k occur. Without a b, coverage at u_{ij} and u_{ik} forces both centers of groups j and k, six centers in all. With one b there is exactly one center in each of the groups j,k; coverage at u_{jk} forces them to have the same color, and then (as $s_j = s_k$) their weighted entries at u_{ij} and u_{ik} are equal. The center b would have to supply the opposite sign at both coordinates, so that $g(i,j) = g(i,k) \neq 0$; but orthogonality of b to v_i gives $g(i,j) + g(i,k) = 0$. This contradiction shows that every group occurs.

If $\mathcal{S}$ contains one center from each group, coverage at u_{12} and u_{34} forces equal colors within $\{1,2\}$ and within $\{3,4\}$. If the two colors were different, all four entries at u_0 would be equal, since the entry of w_i^t there is $t = (ts_i)s_i$. The centers $b^\pm$ vanish at u_0, so $\mathcal{S}$ would not cover $\mathcal{P}$. Hence all four colors are equal. If $\mathcal{S}$ consists of five w's, one group is doubled; coverage at the coordinates joining the three other groups forces their colors to be equal, and the doubled group contains a center of that color. In both cases $\mathcal{S}$ contains

$$F_+ = \{w_1^+, w_2^+, w_3^-, w_4^-\} \quad \text{or} \quad F_- = \{w_1^-, w_2^-, w_3^+, w_4^+\},$$

the four centers of color $+1$ or of color -1. Conversely $F_\pm$ covers $\mathcal{P}$: colors agree on every u_{ij}, and the entries $\pm s_i$ at u_0 are two of each sign. □

C.6 Assembly

Proposition C.8 (The eight-point vanishing)**.** $\widetilde{H}_j(\mathcal{C}_2(8,4);\mathbb{Z}) = 0$ *for* $0 \le j \le 7$.

Proof. By Lemma A.1 with $d = 7$ we must show $\widetilde{H}_t(K_R) = 0$ for $-1 \le t \le 7 - r$, for every face R of nonplanes, where $r = |R|$. For $R = \varnothing$, a set of at most thirteen planes omits some plane p^c and so lies in the simplex of the center p, while every label excludes at least one plane; so $K_\varnothing$ is the boundary of the simplex on $\mathcal{P}$, a 12-sphere, and is acyclic in the required range. For $r \ge 5$ only $t \le 2$ is required, and Lemma C.3 applies. Let $1 \le r \le 4$, put $m = |\operatorname{Ex}(R)|$, and assume that R has an essential center or $m = 14$, since otherwise K_R is contractible (Lemma C.4(a)). By Lemma C.4(b) the required groups are $\widetilde{H}_s(J_R)$ with $s = t + m - 14 \le m - 7 - r$, and groups of degree below -1 vanish. Also $m \le 4r$, and $m \le 11$ when $r = 3$ (Lemma C.2(a)). The cases are as follows.

r	m	required s	reason
1	4	$s \le -4$	nothing to prove
2	≤ 8	$s \le m - 9 \le -1$	J_R has a vertex ($m = 8$)
3	≤ 10	$s \le m - 10 \le 0$	vertex ($m = 9$); connected ($m = 10$)
3	11	$s \le 1$	Lemma C.5
4	≤ 12	$s \le m - 11 \le 1$	vertex, connected, $H_1 = 0$ (Lemma C.4(c))
4	13	$s \le 2$	Lemma C.6(b)
4	14	$t \le 3$ (no shift)	Lemma C.7

The entries not referring to a borderline lemma follow from Lemma C.4(c). It remains to treat the three borderline cases.

Case $r = 3$, $m = 11$. By Lemma C.4(c), J_R is connected, and we must show $H_1(\mathcal{N}'_R) = 0$. Every pair of essential centers is a face of $\mathcal{N}'_R$. A missing triple consists of essential centers Z_1, Z_2, Z_3 whose exclusions cover $\mathcal{P}_1$, so that $|\operatorname{Ex}(Z_1) \cup \operatorname{Ex}(Z_2) \cup \operatorname{Ex}(Z_3)| = 11$. Let M be their majority set. By Lemma C.5(1), M is a center of every label of R. By (2), $\operatorname{Ex}(M) \subseteq \mathcal{P}_1$; a plane center has its exclusion in $\mathcal{P}_0$, so M is an essential center. By (3), M is a cone vertex for the missing triple. Lemma A.3 gives $H_1(\mathcal{N}'_R) = 0$.

Case $r = 4$, $m = 13$. Here $\operatorname{Ex}(R) = \mathcal{P} \smallsetminus \{q\}$ for a plane q, which is the only plane center, $\mathcal{P}_0 = \{q^c\}$ and $\mathcal{P}_1 = \mathcal{P} \smallsetminus \{q^c\}$. By Lemma C.4(c) we must still show $H_2(\mathcal{N}'_R) = 0$, and every triple of essential centers is a face. A missing quadruple would consist of four essential centers, nonplanes whose exclusions cover exactly $\mathcal{P} \smallsetminus \{q^c\}$, and all of whose inner products with the members of R are nonnegative. This contradicts Lemma C.6(b). So $\mathcal{N}'_R$ contains every set of at most four essential centers, and $H_2(\mathcal{N}'_R) = 0$ by Lemma A.3.

Case $r = 4$, $m = 14$. Here $K_R = J_R \simeq \mathcal{N}_R$, and degrees $t \le 2$ are covered by Lemma C.3. By Lemma C.7, the faces of $\mathcal{N}_R$ with at most five elements are exactly the sets of at most five centers containing neither F_+ nor F_-. These are also the faces with at most five elements of the join

$$\partial\Delta(F_+) * \partial\Delta(F_-) * \Delta(\{b^+, b^-\}),$$

whose faces are the sets of centers containing neither F_+ nor F_-. The two complexes have the same four-skeleton and hence the same third homology; the join is a cone with apex b^+. So $H_3(\mathcal{N}_R) = 0$. □

Verification materials

Every finite lemma has a written proof; no theorem depends on computation. The source package contains four supplementary programs and their outputs. Reproduction commands, software

versions, expected outputs, and the scope of each check are given in `Checker_README.txt` in the supplementary source archive. The standard-library program `verify_local_data.py` checks the pentagon and rook incidence identities, the local coloring data in Sections 3 and 8, and both upper-bound constructions. It also checks the Fourier identities and finite classifications in Lemmas C.1, C.2, C.6, and C.7. The standard-library program `check_appendixC_finite.py` adds an exhaustive check of the majority-center Lemma C.5 on all 2,016 extremal triples. The standard-library program `verify_balancing.py` checks the retraction and distance inequality in Theorem 5.1 through ten goods. The optional `check_local_coloring_lemmas.py` checks the two local coloring lemmas using `python-sat` and its Glucose3 solver; it does not produce an independently checked unsatisfiability certificate. None of these programs verifies the homology or chain comparison arguments.

AI methodology statement

We used ChatGPT, including Codex, to assist with literature searches, exposition and proof organization throughout the paper, the figures, and the implementation of `verify_local_data.py` and `verify_balancing.py`. We used Claude to assist with a pre-submission review of the exposition and with the implementation of `check_appendixC_finite.py` and `check_local_coloring_lemmas.py`. The authors verified the correctness and originality of all content, including references.

References

[JA17] A. Jafari and S. Alipour. On the chromatic number of generalized Kneser graphs. *Contributions to Discrete Mathematics* **12**(2) (2017), 69–76. doi:10.55016/ojs/cdm.v12i2.62380.

[BCK19] J. Balogh, D. Cherkashin, and S. Kiselev. Coloring general Kneser graphs and hypergraphs via high-discrepancy hypergraphs. *European Journal of Combinatorics* **79** (2019), 228–236. doi:10.1016/j.ejc.2019.03.004. Published version.

[BK06] E. Babson and D. N. Kozlov. Complexes of graph homomorphisms. *Israel Journal of Mathematics* 152 (2006), 285–312. doi:10.1007/BF02771988. Author's final preprint.

[Bjo95] A. Björner, *Topological methods*, in: Handbook of Combinatorics (R. Graham, M. Grötschel, L. Lovász, eds.), Elsevier, Amsterdam, 1995, 1819–1872.

[BLL+24] X. Bu, Z. Li, S. Liu, J. Song, and B. Tao. Fair division with allocator's preference. In *Web and Internet Economics (WINE 2023)*, Lecture Notes in Computer Science **14413**, Springer (2024), 77–94. doi:10.1007/978-3-031-48974-7_5. Revised full version: arXiv:2310.03475v2 (2025).

[Cso08] P. Csorba. On the simple $\mathbb{Z}_2$-homotopy types of graph complexes and their simple $\mathbb{Z}_2$-universality. *Canadian Mathematical Bulletin* 51(4) (2008), 535–544. doi:10.4153/CMB-2008-053-9.

[DM24] H. R. Daneshpajouh and F. Meunier. Box complexes: at the crossroad of graph theory and topology. arXiv:2307.00299v3, 4 January 2024.

[Dan19] H. R. Daneshpajouh. On the chromatic number of generalized Kneser hypergraphs. *European Journal of Combinatorics* **81** (2019), 150–155. arXiv:1805.11421.

[DF25] M. Dupré la Tour and K. Fujii. *Discrepancy and fair division for non-additive valuations*. arXiv:2509.16802v1, 2025.

[GHI+22] P. W. Goldberg, A. Hollender, A. Igarashi, P. Manurangsi, and W. Suksompong. Consensus halving for sets of items. *Mathematics of Operations Research* **47**(4) (2022), 3357–3379. doi:10.1287/moor.2021.1249. Full version: arXiv:2007.06754.

[GY26] P. Gölz and H. Yaghoubizade. Fair division among couples and small groups. *Proceedings of AAAI 2026* **40** (2026), 16963–16970. Publisher version. Full version: arXiv:2508.13432.

[Hat02] A. Hatcher. *Algebraic Topology*. Cambridge University Press, 2002. Author's online version.

[Hav24] I. Haviv. *Fixed-parameter algorithms for the Kneser and Schrijver problems*. SIAM Journal on Computing **53**(2) (2024), 287–314. doi:10.1137/23M1557076. Full version: arXiv:2204.09009v3; Theorem 1.5 and Remark 6.6 refer to this version.

[HMMS26] A. Hollender, P. Manurangsi, R. Meka, and W. Suksompong. *Discrepancy beyond additive functions with applications to fair division (Extended Abstract)*. In ITCS 2026, LIPIcs **362**, 77:1. doi:10.4230/LIPIcs.ITCS.2026.77. Full version: arXiv:2509.09252v2; result numbering refers to this version.

[KSV20] M. Kyropoulou, W. Suksompong, and A. A. Voudouris. Almost envy-freeness in group resource allocation. *Theoretical Computer Science* **841** (2020), 110–123. doi:10.1016/j.tcs.2020.07.008.

[Lov78] L. Lovász, *Kneser's conjecture, chromatic number, and homotopy*, J. Combin. Theory Ser. A **25** (1978), 319–324.

[Mat03] J. Matoušek. *Using the Borsuk–Ulam Theorem: Lectures on Topological Methods in Combinatorics and Geometry*. Springer, 2003; corrected second printing, 2008. All page references are to the printed pagination of this printing.

[MZ04] J. Matoušek and G. M. Ziegler. Topological lower bounds for the chromatic number: a hierarchy. *Jahresbericht der Deutschen Mathematiker-Vereinigung* **106** (2004), 71–90. arXiv:math/0208072.

[RS72] C. P. Rourke and B. J. Sanderson, *Introduction to Piecewise-Linear Topology*, Springer, 1972.

[ST06] G. Simonyi and G. Tardos. Local chromatic number, Ky Fan's theorem, and circular colorings. *Combinatorica* **26**(5) (2006), 587–626. doi:10.1007/s00493-006-0034-x. Full version: arXiv:math/0407075.

[Spa66] E. H. Spanier, *Algebraic Topology*, McGraw-Hill, 1966.

Sharareh Alipour: `sh.alipour@newuu.uz`
Amir Jafari: `a.jafari@newuu.uz`